

Robust Replacement of Human Decision Makers under Distribution Shift

Dingli Wang*

Abstract

We study when a machine should replace a human before its deployment distribution is known. A robust receiver operating characteristic frontier evaluates fixed thresholds over a label-preserving Wasserstein set, while historical decisions give a posterior for human performance, assumed stable across environments. Replacement requires a specified posterior probability of no higher false-positive rate and no lower true-positive rate throughout the set. We establish uniform frontier estimation and welfare bounds that separate whom to replace from which threshold to deploy. Threshold choice can leave a persistent welfare difference from a deployment benchmark that matches the human’s false-positive rate, even with a known frontier and increasingly informative histories. For logistic scores, average and per-observation displacement budgets affect threshold probabilities at square-root and linear local orders, respectively. Simulations holding the score fixed show partial replacement and positive mean gains at evaluated distributions within the per-observation set, while the corresponding average budgets yield retention. Choosing thresholds by a welfare lower bound within the same posterior feasible set improves mean deployment welfare in this design.

Keywords: Distributionally robust optimization, Wasserstein distance, ROC curve, human-algorithm comparison, distribution shift.

JEL Classification: C11, C12, C44, C45

*Department of Economics, University of California, Riverside, CA 92521, USA.
E-mail: dwang177@ucr.edu.

1 Introduction

A more accurate prediction is not enough to justify replacing a human decision maker. The decision also depends on which errors matter, which machine threshold will be used, and whether the comparison will survive a change in the environment. An algorithm may outperform a doctor on historical cases yet perform differently when the recorded inputs change. A decision maker who must choose a replacement before observing that change faces a different problem from one who can select a threshold after seeing the deployment population.

The human–algorithm literature provides methods for making individual replacement decisions rather than comparing only average human and machine performance (Kleinberg et al., 2018; Mullainathan and Obermeyer, 2022; Feng et al., 2025). In the ROC approach of Feng et al. (2025), the relevant objects are the human’s false-positive rate (FPR), true-positive rate (TPR), and the machine’s receiver operating characteristic (ROC) curve. A machine point dominates a human if it has no higher FPR and no lower TPR. Their Bayesian rule selects the point with the largest posterior probability of dominance and replaces the human when that probability exceeds a specified threshold. It therefore accounts for uncertainty about an individual human’s performance.

This paper adds uncertainty about the environment in which that fixed machine threshold will operate. Historical decisions support a posterior for human performance, whereas the future deployment distribution is described by a set of possible laws. We study a setting in which the human’s class-conditional error rates remain stable but the machine’s recorded predictors may change. For example, clinical assessment may continue to use the same diagnostic information while a change in recording or measurement affects the variables supplied to an algorithm. This setting distinguishes uncertainty about an individual’s performance from uncertainty about the machine’s future inputs.

We construct a robust ROC frontier by evaluating each threshold before comparing it with the human. For that threshold, we compute the largest FPR and the smallest TPR over a Wasserstein neighborhood of the reference distribution. We apply the fixed-threshold posterior dominance rule to the resulting frontier. The decision maker prefers replacement when a posterior requirement of no deterioration in either rate is met, and otherwise retains the human. The posterior requirement supplies the safety constraint; a second preference determines which feasible threshold is deployed. The transport model preserves the label proportions and allows class-conditional predictors to change; it is distinct from classical covariate shift, which keeps the label distribution conditional on the predictors unchanged (Shimodaira, 2000; Sugiyama et al., 2007).

The welfare analysis separates two decisions: whom to replace and which machine threshold to deploy. We bound harm relative to retaining the human, using errors in the estimated robust performance and in the human posterior. We also compare the

implemented policy with a benchmark that knows the deployment ROC and matches each human’s FPR. A margin restriction limits the losses attributable to differences in the replacement set. Even when two policies replace exactly the same humans, however, they can choose different thresholds and obtain different welfare. The bound retains this contribution. An analytical example shows that, even with a known machine frontier and increasingly informative human histories, the posterior-maximizing threshold can remain separated from the deployment matched-FPR benchmark. Thus evidence that a machine improves both rates does not determine the welfare obtained from its chosen threshold.

We establish uniform estimation of the threshold probabilities and frontier under function-class and local FPR conditions, with a worked logistic-score example. Additional conditions give a functional limit. An extension bounds each observation’s displacement. For a logistic score, it permits exact local extrema and corresponding replacement and welfare bounds. Average and pointwise budgets affect threshold probabilities at different local orders. For finite tree scores, statistical approximation and the finite search of local extrema are separate issues.

The numerical evidence distinguishes the size of the budget from the restriction it imposes. Nominal replacement improves welfare near the reference distribution but incurs losses after larger shifts in recorded inputs. The Wasserstein budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$ retain every doctor in this design; smaller budgets give partial replacement. With the same fitted logistic score, a pointwise radius of 0.05 replaces 18.68% and yields a positive mean welfare gain over retaining everyone at the evaluated boundary of that set. The smaller pointwise set excludes the concentration of displacement allowed by the average budget. On the same posterior feasible sets, welfare-based threshold choice raises mean welfare in the evaluated design while accepting lower posterior dominance probabilities. An auxiliary cross-state income illustration uses observed predictors and synthetic human decisions.

The closest starting point is the individual ROC-based rule of [Feng et al. \(2025\)](#). We connect its fixed-threshold decision to distributional uncertainty and distinguish replacement selection from threshold choice in welfare. The probability calculations use optimal-transport risk bounds ([Blanchet and Murthy, 2019](#); [Gao and Kleywegt, 2023](#); [Mohajerin Esfahani and Kuhn, 2018](#)). In particular, [Shafieezadeh-Abadeh et al. \(2015, Section 3.3, Theorem 3\)](#) and [Shafieezadeh-Abadeh et al. \(2019, Section 5, Theorem 42\)](#) already derive worst- and best-case misclassification bounds for a fixed linear classifier. The logistic case of [Proposition 1](#) uses the same distance-to-half-space geometry after conditioning on the label, with the strict-threshold conventions stated there. We use these class-specific probabilities to compare the machine with each human’s posterior and to analyze the welfare of a threshold chosen before deployment.

More broadly, [Kuhn et al. \(2019\)](#) review Wasserstein distributionally robust optimization for learning and decision making. [Duchi and Namkoong \(2021\)](#) study uniform

performance using f -divergence neighborhoods, a different restriction on distributional uncertainty. The margin argument is related to plug-in classification (Mammen and Tsybakov, 1999; Audibert and Tsybakov, 2007), here with an explicitly fixed group of decision makers. The analysis also complements work on binary decisions and ROC inference (Elliott and Lieli, 2013; Feng et al., 2026). Importance weighting evaluates a specified target law using a density ratio (Sugiyama and Kawanabe, 2012); our threshold is chosen before that law is known.

Sections 2–3 present the decision rule and its statistical and welfare properties. Section 4 gives the local-perturbation extension; Sections 5–6 report the numerical evidence. Section 7 concludes. Proofs and additional inference results are in Appendix A.

2 The replacement decision under distribution shift

2.1 Information and human performance

We begin with the ROC comparison in Feng et al. (2025). Consider J human decision makers, indexed by $j = 1, \dots, J$, whom we call doctors in the motivating example. Let P_0 be the reference law of (X, Y) . The validation observations and the historical records $\mathcal{D}_j$ used to estimate doctor j 's performance come from this reference environment. The decision maker observes these data, then decides whether to retain each doctor or replace that doctor with a machine using a single threshold. The threshold is chosen before the deployment law is realized and is not adjusted to that law.

The machine score is trained on a separate sample. We condition on that training sample when analyzing the validation estimator, so m is a fixed function in the analysis. The regularity conditions below apply to this fitted score. For observation i assigned to decision maker j , $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ are covariates, $Y_i \in \{0, 1\}$ is the ground-truth label, $\hat{Y}_{i,j} \in \{0, 1\}$ is the human decision, and $\hat{Y}_{i,M} \in \{0, 1\}$ is the machine decision. The human's false-positive and true-positive rates are

$$\alpha_{H,j} = \frac{\mathbb{E}[(1 - Y_i)\hat{Y}_{i,j}]}{1 - p}, \quad \beta_{H,j} = \frac{\mathbb{E}[Y_i\hat{Y}_{i,j}]}{p}, \quad (2.1)$$

where $p = p_0 = P_0(Y = 1)$ is the reference prevalence and the expectations in (2.1) are taken under the reference human decision process. We write $\theta_{H,j} = (\alpha_{H,j}, \beta_{H,j})$.

The machine produces a score $m(x) \in [0, 1]$ and uses the strict threshold rule $\hat{Y}_{i,M} = \mathbb{I}(m(X_i) > c)$, where $\mathbb{I}(A)$ is the indicator of event A . We write the ROC frontier using an FPR constraint:

$$g(\alpha) = \sup_{c \in [0,1]} \left\{ \frac{\mathbb{E}[Y \cdot \mathbb{I}(m(X) > c)]}{p} \mid \frac{\mathbb{E}[(1 - Y) \cdot \mathbb{I}(m(X) > c)]}{1 - p} \leq \alpha \right\}. \quad (2.2)$$

Using an inequality accommodates unattainable FPR values when scores have atoms, while retaining deterministic thresholds. When the negative-class score distribution is continuous and strictly increasing over the thresholds of interest, the constraint binds and this definition agrees with the matched-FPR ROC. If $m(x) = \mathbb{P}(Y = 1 \mid X = x)$ and the class-conditional score laws are continuous, thresholding yields the Neyman–Pearson ROC based on X .

Feng et al. (2025) replace human j if and only if

$$\hat{q}_{\max,j} := \max_{\alpha \in [0,1]} \Pi_j \{ \alpha_H \geq \alpha, \beta_H \leq g(\alpha) \mid \mathcal{D}_j \} \geq \tau, \quad (2.3)$$

Here $\Pi_j(\cdot \mid \mathcal{D}_j)$ is the Dirichlet–multinomial posterior for doctor j ’s rate pair; $\theta_H = (\alpha_H, \beta_H)$ denotes the pair integrated over in that posterior, while $\theta_{H,j}$ denotes its true value. The cutoff $\tau \in (0, 1)$ is the required posterior probability of dominance.

Under a deployment law Q , write $\text{FPR}_Q(c) = Q(m(X) > c \mid Y = 0)$ and $\text{TPR}_Q(c) = Q(m(X) > c \mid Y = 1)$. We write the frontier as $g(\alpha; P)$ when its reference law matters, and as $g_Q(\alpha) := g(\alpha; Q)$ under Q .

Human performance across environments. A deployment environment specifies both a law of (X, Y) and a human decision process. The ambiguity set below restricts the (X, Y) marginal. For the main comparison, we maintain that every admitted environment has the same human class-conditional error rates as the reference environment:

$$\mathbb{P}_Q(\hat{Y}_j = 1 \mid Y = 0) = \alpha_{H,j}, \quad \mathbb{P}_Q(\hat{Y}_j = 1 \mid Y = 1) = \beta_{H,j}. \quad (2.4)$$

Here $\mathbb{P}_Q$ refers to the full environment, including the human decision process, with (X, Y) marginal Q . The historical posterior therefore concerns the rate pair used in deployment.

One sufficient environment has a doctor choosing $h_j(U_j)$ from a diagnostic signal U_j , with h_j and the law of $U_j \mid Y$ stable across environments, while the machine’s recorded predictors $X \mid Y$ may change. Then $\alpha_{H,j}$ and $\beta_{H,j}$ are integrals of the fixed rule h_j under two stable signal laws. Equation (2.4) is a restriction on the environments being compared; it does not follow from the transport restriction on (X, Y) .

Historical decisions provide a likelihood for the fixed human rate pair; combining it with the prior in Feng et al. (2025) gives the posterior. The decision maker specifies possible future machine-input distributions without assigning probabilities to them and requires the comparison to hold throughout that set. Allowing human performance to change would also require specifying the human decision process in each environment. Under (2.4), the historical posterior estimates the common human rates, while validation data estimate the machine frontier under the reference law.

2.2 Wasserstein ambiguity and the thresholdwise robust frontier

Let $\mathcal{Z} := \mathcal{X} \times \{0, 1\}$ and let $\hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ be the empirical distribution of n validation observations, where δ_z denotes a unit point mass at z . For a transport cost $\kappa : \mathcal{Z} \times \mathcal{Z} \rightarrow [0, \infty]$, define

$$\mathcal{W}_\kappa(P, Q) = \inf_{\pi \in \Pi(P, Q)} \int \kappa(z, z') d\pi(z, z'), \quad (2.5)$$

where $\Pi(P, Q)$ is the set of couplings with marginals P and Q .

We use a label-preserving transport cost:

$$\kappa((x, y), (x', y')) = \begin{cases} \|x - x'\|_2, & y = y', \\ +\infty, & y \neq y'. \end{cases} \quad (2.6)$$

Write $\mathcal{P}(\mathcal{Z})$ for the Borel probability laws on $\mathcal{Z}$. For any reference law P , define

$$\mathcal{B}_\varepsilon(P) = \{Q \in \mathcal{P}(\mathcal{Z}) : \mathcal{W}_\kappa(Q, P) \leq \varepsilon\}. \quad (2.7)$$

Finite transport cost requires $Q(Y = 1) = P(Y = 1)$. Write P^y and Q^y for the conditional laws of X given $Y = y$, and let W_1 use the Euclidean ground metric. With $p = P(Y = 1) \in (0, 1)$, the cost separates by class:

$$\mathcal{W}_\kappa(P, Q) = pW_1(P^1, Q^1) + (1 - p)W_1(P^0, Q^0). \quad (2.8)$$

Thus the model allows class-conditional feature distributions to change while preserving prevalence. It does not impose $Q(Y | X) = P(Y | X)$, the conditional invariance used in classical covariate shift (Sugiyama et al., 2007). The same transport budget is available separately when computing a worst-case TPR and a worst-case FPR; a single law need not attain both extremes.

Definition 1 (Thresholdwise robust frontier). *Let P be a distribution on (X, Y) with $p := P(Y = 1) \in (0, 1)$. For each threshold $c \in [0, 1]$, define*

$$\underline{\beta}_\varepsilon(c; P) := \inf_{Q \in \mathcal{B}_\varepsilon(P)} Q(m(X) > c | Y = 1),$$

$$\overline{\alpha}_\varepsilon(c; P) := \sup_{Q \in \mathcal{B}_\varepsilon(P)} Q(m(X) > c | Y = 0).$$

The robust ROC frontier is

$$g_\varepsilon(\alpha; P) := \sup \left\{ \underline{\beta}_\varepsilon(c; P) : \overline{\alpha}_\varepsilon(c; P) \leq \alpha \right\}, \quad \alpha \in [0, 1]. \quad (2.9)$$

A threshold is feasible at α when its FPR is at most that level throughout the

deployment set. The frontier gives the largest guaranteed TPR achievable with one such threshold chosen before deployment.

Proposition 1 (Fixed-threshold exact dual for robust TPR/FPR). *Let $\mathcal{X}$ be a closed subset of $\mathbb{R}^d$, let P have a finite first moment and $p \in (0, 1)$, and let m be continuous. Fix $\varepsilon > 0$ and a threshold c for which $S_c := \{x \in \mathcal{X} : m(x) > c\}$ and its complement are both nonempty. Define distances within $\mathcal{X}$ by*

$$d_c^-(x) := \inf\{\|x - z\|_2 : m(z) \leq c\}, \quad d_c^+(x) := \inf\{\|x - z\|_2 : m(z) > c\}.$$

Then

$$\underline{\beta}_\varepsilon(c; P) = \sup_{\lambda \geq 0} \left\{ \frac{1}{p} \mathbb{E}_P[Y \min\{1, \lambda d_c^-(X)\}] - \frac{\lambda}{p} \varepsilon \right\}, \quad (2.10)$$

and

$$\overline{\alpha}_\varepsilon(c; P) = \inf_{\lambda \geq 0} \left\{ \frac{\lambda}{1-p} \varepsilon + \frac{1}{1-p} \mathbb{E}_P[(1-Y)(1 - \lambda d_c^+(X))_+] \right\}. \quad (2.11)$$

Both optimizations can be restricted to $0 \leq \lambda \leq 1/\varepsilon$. The same interval applies when P is replaced by an empirical law containing both classes. If S_c is empty, both rates are zero; if $S_c = \mathcal{X}$, both rates are one.

Here $(u)_+ = \max\{u, 0\}$ denotes the positive part. The proof uses the closed-event form of transport duality in [Blanchet and Murthy \(2019, Theorem 1 and Section 2.4\)](#) after conditioning on the label. For the open event S_c , it additionally proves that, at positive radius, maximizing its probability and maximizing the probability of its closure have the same value. This is an equality of suprema; it need not imply existence of a worst-case law for the open event.

At $\varepsilon = 0$ we use the nominal strict-threshold probabilities

$$\underline{\beta}_0(c; P) = P^1(S_c), \quad \overline{\alpha}_0(c; P) = P^0(S_c).$$

The zero-radius limit of the FPR dual can instead count the closure of S_c . It agrees with the nominal FPR if $P^0(\overline{S_c} \setminus S_c) = 0$, but this condition is not imposed on empirical thresholds. Empty decision regions are evaluated directly rather than through products involving an infinite distance.

2.3 The robust replacement decision

Definition 2 (Robust dominance). *A point $\theta_M = (\alpha_M, \beta_M)$ on the robust ROC curve $g_\varepsilon(\cdot)$ robustly dominates a human FPR/TPR pair $\theta_H = (\alpha_H, \beta_H)$ if $\alpha_M \leq \alpha_H$ and $\beta_M \geq \beta_H$. Thus θ_H is dominated by some frontier point if and only if $\beta_H \leq g_\varepsilon(\alpha_H)$.*

For a fixed threshold c , the event

$$\bar{\alpha}_\varepsilon(c; P) \leq \alpha_H, \quad \underline{\beta}_\varepsilon(c; P) \geq \beta_H$$

means that this one threshold weakly dominates the human pair for every $Q \in \mathcal{B}_\varepsilon(P)$, under the maintained human comparison in (2.4). Define its posterior probability by

$$q_j^\varepsilon(c; P) := \Pi_j(\alpha_H \geq \bar{\alpha}_\varepsilon(c; P), \beta_H \leq \underline{\beta}_\varepsilon(c; P) \mid \mathcal{D}_j). \quad (2.12)$$

The probability is over the human pair; the same threshold must dominate it throughout the deployment set.

Definition 3 (Robust Bayesian replacement). *Given $g_\varepsilon(\cdot; P)$ and the doctor posterior, define*

$$\hat{q}_{\max, j}^\varepsilon := \max_{\alpha \in [0, 1]} \Pi_j(\alpha_H \geq \alpha, \beta_H \leq g_\varepsilon(\alpha; P) \mid \mathcal{D}_j). \quad (2.13)$$

The rule is defined when the displayed maximum is attained. Replace doctor j if $\hat{q}_{\max, j}^\varepsilon \geq \tau$, using a threshold implementing a maximizing frontier point, and retain the doctor if $\hat{q}_{\max, j}^\varepsilon < \tau$. Finite candidate sets ensure attainment.

Lemma 2 shows that the frontier is attained at its smallest feasible threshold and that

$$\sup_{\alpha \in [0, 1]} \Pi_j\{\alpha_H \geq \alpha, \beta_H \leq g_\varepsilon(\alpha; P) \mid \mathcal{D}_j\} = \sup_{c \in [0, 1]} q_j^\varepsilon(c; P).$$

Thus, whenever the maximum exists, the threshold selected by Definition 3 can also be selected by maximizing (2.12). The result also holds for exact empirical components and, using its own frontier, for a finite threshold grid containing $c = 1$.

The decision maker values freeing human capacity, subject to a required posterior probability of no higher FPR and no lower TPR. With replacement indicator $r_j \in \{0, 1\}$, the first decision is

$$\max_{r_j \in \{0, 1\}, c \in [0, 1]} r_j \quad \text{subject to} \quad r_j \{1 - q_j^\varepsilon(c; P)\} \leq 1 - \tau. \quad (2.14)$$

Retention is always feasible. A replacement is feasible when at least one threshold satisfies the posterior requirement. Among feasible thresholds, Definition 3 gives priority to the probability of dominance. The second preference determines the deployed threshold.

A decision maker who specifies a value $\phi \geq 0$ for true positives and a cost $\eta \geq 0$ for false positives can instead maximize a welfare lower bound within the same feasible set. To state the two choices for the estimated components, let $\mathcal{C}_n$ be the candidate set and

write $\widehat{q}_j^\varepsilon(c) = q_j^\varepsilon(c; \widehat{\mathbb{P}}_n)$. Define

$$\begin{aligned}\mathcal{C}_j^\tau &= \{c \in \mathcal{C}_n : \widehat{q}_j^\varepsilon(c) \geq \tau\}, \\ \widehat{V}_\varepsilon(c) &= \phi \underline{\beta}_\varepsilon(c; \widehat{\mathbb{P}}_n) - \eta \overline{\alpha}_\varepsilon(c; \widehat{\mathbb{P}}_n), \\ \widehat{c}_j^q &\in \arg \max_{c \in \mathcal{C}_n} \widehat{q}_j^\varepsilon(c), \quad \widehat{c}_j^W \in \arg \max_{c \in \mathcal{C}_j^\tau} \widehat{V}_\varepsilon(c).\end{aligned}\tag{2.15}$$

The second choice is used when $\mathcal{C}_j^\tau$ is nonempty; otherwise both rules retain the human. We use measurable maximizers, with the smallest threshold breaking exact ties on a finite grid. Both rules replace the same humans. For each replaced human,

$$\widehat{q}_j^\varepsilon(\widehat{c}_j^q) \geq \widehat{q}_j^\varepsilon(\widehat{c}_j^W) \geq \tau, \quad \widehat{V}_\varepsilon(\widehat{c}_j^W) \geq \widehat{V}_\varepsilon(\widehat{c}_j^q).\tag{2.16}$$

We retain $\widehat{c}_j^q$ as the principal rule because it does not require relative error costs. The alternative uses these costs once the posterior requirement is met. Under the average transport budget, $\widehat{V}_\varepsilon$ is a lower bound formed from the two worst-case rates. For exact logistic pointwise perturbations (Section 4.1), it equals worst-case welfare over the set centered at the empirical distribution. Section 3.3 includes estimation error in the population comparison.

At the population frontier, (2.12) is an exact posterior probability of the stated event. In implementation we substitute the validation law $\widehat{\mathbb{P}}_n$ for P and record the chosen threshold $\widehat{c}_j$. Sampling error in these substituted components is separate from posterior uncertainty about the human pair. Appendix A.10 gives the corresponding rule when simultaneous confidence bounds on the components are available. Figure 1 shows how a larger radius lowers the frontier available for comparison. Since $g_\varepsilon \leq g$, robust replacement is more selective than nominal replacement at the same posterior threshold.

Proposition 2 (Conservativeness of robust replacement). *For any $\varepsilon > 0$, the set of robustly replaced doctors is a subset of the set of non-robustly replaced doctors:*

$$\{j : \widehat{q}_{\max,j}^\varepsilon \geq \tau\} \subseteq \{j : \widehat{q}_{\max,j}^0 \geq \tau\}.$$

The radius ε specifies the deployment laws against which replacement is assessed. Observed environmental distances can inform its scale only when they use the same ground metric and reference center as the ambiguity set. Under label-preserving transport, two empirical laws with different class proportions have infinite joint distance. A marginal feature distance can describe their covariate differences, but its median or maximum does not establish membership in the joint ambiguity set. A specified radius grid instead reports how the decision changes with the scope of protection.

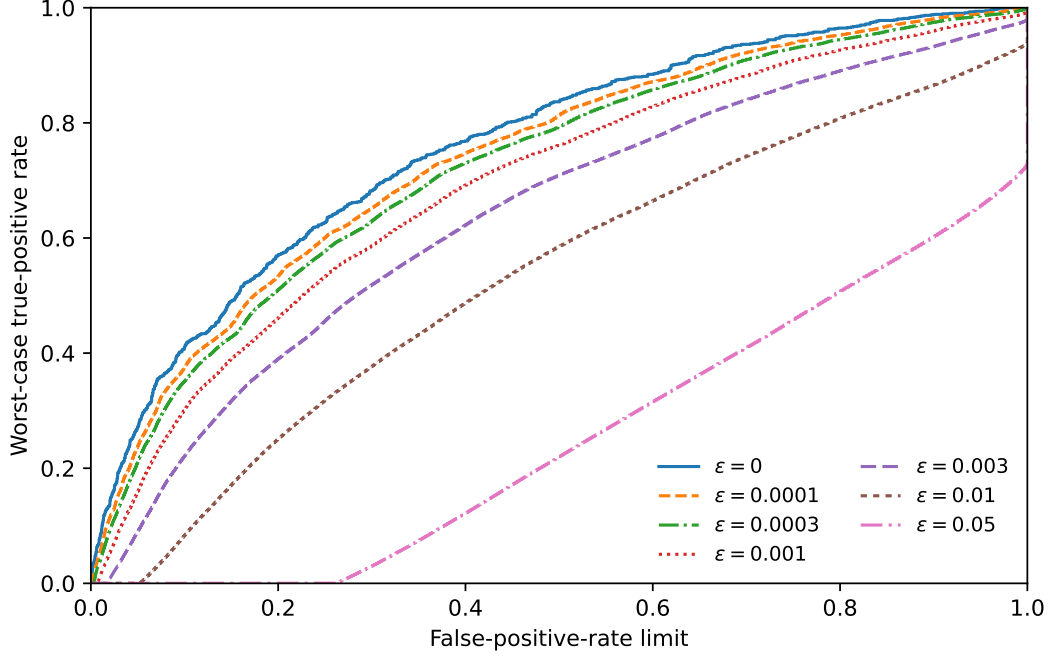


Figure 1: How the deployment radius changes the robust ROC frontier.

Notes: Frontiers for replication 0 in Section 5, using its fitted score, validation records and 1,001 candidate thresholds. Each curve gives the largest lower TPR bound subject to the upper FPR limit on the horizontal axis. Steps correspond to actual feasible candidates. All seven radii in the finer grid are shown.

2.4 Computation

Write $\hat{\underline{\beta}}_\varepsilon(c) = \underline{\beta}_\varepsilon(c; \hat{\mathbb{P}}_n)$ and $\hat{\underline{\alpha}}_\varepsilon(c) = \underline{\alpha}_\varepsilon(c; \hat{\mathbb{P}}_n)$ for the empirical robust components. The empirical frontier is obtained from these components by the same FPR constraint.

For doctor j , let $(\text{TP}_j, \text{FP}_j, \text{FN}_j, \text{TN}_j)$ count historical true positives, false positives, false negatives and true negatives. Under the symmetric Dirichlet prior with parameter $a_0 > 0$ on each cell, the posterior rates are independent and satisfy

$$\begin{aligned} \alpha_H \mid \mathcal{D}_j &\sim \text{Beta}(a_0 + \text{FP}_j, a_0 + \text{TN}_j), \\ \beta_H \mid \mathcal{D}_j &\sim \text{Beta}(a_0 + \text{TP}_j, a_0 + \text{FN}_j). \end{aligned} \quad (2.17)$$

This is the confusion-table posterior of Feng et al. (2025). Let $F_{\alpha,j}$ and $F_{\beta,j}$ denote the posterior cumulative distribution functions (CDFs) and $S_{\alpha,j} = 1 - F_{\alpha,j}$. With empirical robust components, the fixed-threshold posterior probability is

$$\hat{q}_j^\varepsilon(c) = S_{\alpha,j}(\hat{\underline{\alpha}}_\varepsilon(c)) F_{\beta,j}(\hat{\underline{\beta}}_\varepsilon(c)). \quad (2.18)$$

The Beta laws are continuous, so these CDF expressions respect the weak dominance inequalities. Lemma 6 establishes the factorization and its concentration properties.

On a finite candidate grid containing $c = 1$, maximize (2.18) with a predetermined

tie-breaking rule. Replace the doctor if the maximum is at least τ , deploying that same threshold. The Beta formula removes posterior-sampling error; component estimation, numerical evaluation, and grid approximation remain separate. At zero radius, use the empirical strict-threshold rates. Very small probabilities, or failure probabilities close to zero, are compared in the log domain.

For positive radius, each dual objective evaluation requires $O(n)$ operations once the distances $d_c^\pm(X_i)$ have been computed. Total computation also includes obtaining the distances and solving the scalar optimization.

An exact calculation for a logistic score. For $m(x) = \{1 + \exp[-(b + x^\top \beta)]\}^{-1}$ on $\mathcal{X} = \mathbb{R}^d$, with $\beta \neq 0$, define $z(x) = b + x^\top \beta$ and $k_c = \log\{c/(1 - c)\}$ for $0 < c < 1$. The distances used in Proposition 1 are

$$d_c^-(x) = \frac{(z(x) - k_c)_+}{\|\beta\|_2}, \quad d_c^+(x) = \frac{(k_c - z(x))_+}{\|\beta\|_2}. \quad (2.19)$$

These are distances to half-spaces in the covariate metric. This is the boundary geometry used in fixed-classifier risk bounds by Shafieezadeh-Abadeh et al. (2015, Section 3.3, Theorem 3) and Shafieezadeh-Abadeh et al. (2019, Section 5, Theorem 42). Here it is applied separately within the two label classes. Restricted covariate supports require distances calculated within that support.

At positive radius, the empirical dual is piecewise linear in λ . It can be evaluated at $0, 1/\varepsilon$, and the reciprocal positive distances that fall in $[0, 1/\varepsilon]$. Equivalently, sort the distances within each class and allocate transport to the closest observations first, allowing a fraction of the last observation's mass to move. Each observation has joint mass $1/n$, so its full transport cost is d_i/n ; the total budget is ε , not a separate conditional budget of the same size in each class. Lemma 3 establishes this calculation. Insert these components into (2.18), using the zero-radius and constant-rule conventions in Proposition 1.

Remark 1 (The scale of the average displacement budget). *For the logistic score in (2.19), let $S = z(X)/\|\beta\|_2$ and $a_c = k_c/\|\beta\|_2$. Write H_y for the conditional CDF of $S \mid Y = y$, and let $\alpha_0(c) = \bar{\alpha}_0(c; P)$ and $\beta_0(c) = \underline{\beta}_0(c; P)$ denote the nominal rates. At a fixed interior threshold where both CDFs have continuous positive densities, as $\varepsilon \downarrow 0$,*

$$\bar{\alpha}_\varepsilon(c) - \alpha_0(c) = \sqrt{\frac{2\varepsilon H'_0(a_c)}{1 - p}} + o(\sqrt{\varepsilon}), \quad (2.20)$$

$$\beta_0(c) - \underline{\beta}_\varepsilon(c) = \sqrt{\frac{2\varepsilon H'_1(a_c)}{p}} + o(\sqrt{\varepsilon}). \quad (2.21)$$

Moving the closest mass across the boundary minimizes its transport cost. A strip of width w has mass $H'_y(a_c)w + o(w)$ and cost $H'_y(a_c)w^2/2 + o(w^2)$ within its class. Equating this

cost to ε/p or $\varepsilon/(1-p)$ gives the expansions. For the strict positive event, movement just beyond the boundary yields the same supremal value. Example 1 gives the exact uniform-density calculation $s = 2\sqrt{\varepsilon}$.

An average budget can therefore move substantially more probability across a threshold than a uniform translation of the same numerical size. These are population local expansions at a fixed threshold. In a fixed empirical sample, probability changes are piecewise linear in the positive budget. The pointwise budget in Section 4.1 limits each observation separately and has a different local scale.

3 Statistical Theory

We first control estimation of the threshold probabilities and frontier, then analyze the implemented replacement rule and its welfare consequences.

3.1 Estimating the robust frontier

Continue to write $p_0 = P_0(Y = 1)$. Let $\hat{p} := n^{-1} \sum_{i=1}^n \mathbb{I}(Y_i = 1)$ denote the empirical class prevalence. For $\varepsilon > 0$, on samples with $0 < \hat{p} < 1$ and at thresholds with two nonempty decision regions, Proposition 1 gives the sample robust TPR component

$$\hat{\beta}_{\varepsilon}(c) := \sup_{\lambda \geq 0} \left\{ \frac{1}{\hat{p}} \frac{1}{n} \sum_{i=1}^n Y_i \min\{1, \lambda d_c^-(X_i)\} - \frac{\lambda}{\hat{p}} \varepsilon \right\}, \quad (3.1)$$

and the sample robust FPR component is

$$\hat{\alpha}_{\varepsilon}(c) := \inf_{\lambda \geq 0} \left\{ \frac{\lambda}{1 - \hat{p}} \varepsilon + \frac{1}{1 - \hat{p}} \frac{1}{n} \sum_{i=1}^n (1 - Y_i) (1 - \lambda d_c^+(X_i))_+ \right\}. \quad (3.2)$$

Constant threshold rules are evaluated directly, and at zero radius the components are the nominal empirical rates with the same strict inequality. The sample thresholdwise robust frontier is then

$$\hat{g}_{\varepsilon}(\alpha) := \sup \left\{ \hat{\beta}_{\varepsilon}(c) : \hat{\alpha}_{\varepsilon}(c) \leq \alpha \right\}.$$

Define the exact-dual functions:

$$\begin{aligned} \phi_{\lambda,c}^+(x, y) &:= \frac{y}{p_0} \min\{1, \lambda d_c^-(x)\}, \\ \phi_{\lambda,c}^-(x, y) &:= \frac{1-y}{1-p_0} (1 - \lambda d_c^+(x))_+. \end{aligned}$$

Uniform estimation requires control of the function classes and a stable inverse of the FPR map. The next assumption states these requirements, using the common

multiplier interval from Proposition 1. Its covering bound is the usual polynomial bound for Vapnik–Chervonenkis (VC)-type classes.

Assumption 1 (Complexity and regularity of the thresholdwise robust frontier). *Conditional on the independent training sample, validation observations are independent with common law P_0 satisfying the conditions of Proposition 1. Fix $\varepsilon > 0$ and a compact interval $\mathcal{C} = [c_l, c_u] \subset (0, 1)$ with nonempty decision regions throughout. The following conditions hold:*

- (i) *population and empirical dual optimizations use $\Lambda = [0, 1/\varepsilon]$;*
- (ii) *the classes $\mathcal{F}^\pm := \{\phi_{\lambda, c}^\pm : (\lambda, c) \in \Lambda \times \mathcal{C}\}$ are pointwise measurable and have bounded envelopes. For each class there are fixed finite constants K, v and a constant envelope M such that*

$$\sup_H N(uM, \mathcal{F}^\pm, L_2(H)) \leq (K/u)^v, \quad 0 < u < 1,$$

where H ranges over finitely supported probability laws and $N(t, \mathcal{F}, L_2(H))$ is the minimum number of $L_2(H)$ balls of radius t covering $\mathcal{F}$. Pointwise measurability means that a countable subclass approximates every class member pointwise;

- (iii) $0 < p_{\min} \leq p_0 \leq p_{\max} < 1$;

- (iv) *on $\mathcal{C}$, the population components are continuously differentiable, with*

$$0 < \underline{\kappa} \leq -\partial_c \bar{\alpha}_\varepsilon(c; P_0) \leq \bar{\kappa} < \infty, \quad |\partial_c \beta_\varepsilon(c; P_0)| \leq L_\beta < \infty.$$

For the interval $[\alpha_l, \alpha_u] \subset (0, 1)$ under consideration, there is $b_C > 0$ such that

$$\bar{\alpha}_\varepsilon(c_l; P_0) > \alpha_u + b_C, \quad \bar{\alpha}_\varepsilon(c_u; P_0) < \alpha_l - b_C.$$

Thus the FPR interval lies strictly inside the range of the regular threshold region.

Remark 2 (A verifiable complexity condition for logistic scores). *For the logistic score on $\mathbb{R}^d$ in (2.19), the exact-dual functions are clipped affine functions of the fixed index $z(x)$. Lemma 3 verifies their VC-subgraph complexity and pointwise measurability, establishing part (ii). Part (iv) additionally requires the stated derivative bounds and an FPR interval inside the range of the strictly decreasing FPR component.*

For this interval, write

$$c_\alpha = \min\{c \in \mathcal{C} : \bar{\alpha}_\varepsilon(c; P_0) \leq \alpha\}, \quad \mathcal{C}_\alpha = \{c_\alpha : \alpha \in [\alpha_l, \alpha_u]\}.$$

The endpoint inequalities keep these inverse thresholds, and their empirical counterparts, within $\mathcal{C}$ when component error is small. The proof allows jumps in the empirical FPR.

Theorem 1 (Uniform convergence of the thresholdwise robust frontier). *Under Assumption 1, for fixed $\varepsilon > 0$:*

(i)

$$\sup_{c \in \mathcal{C}} \left| \widehat{\beta}_{\underline{\varepsilon}}(c) - \underline{\beta}_{\varepsilon}(c; P_0) \right| + \sup_{c \in \mathcal{C}} \left| \widehat{\alpha}_{\varepsilon}(c) - \overline{\alpha}_{\varepsilon}(c; P_0) \right| = O_p(n^{-1/2});$$

(ii) on the FPR interval with the interior-range condition in Assumption 1(iv),

$$\sup_{\alpha \in [\alpha_l, \alpha_u]} |\widehat{g}_{\varepsilon}(\alpha) - g_{\varepsilon}(\alpha; P_0)| = O_p(n^{-1/2}). \quad (3.3)$$

The replacement and welfare results use this uniform accuracy. Appendix A.5 gives a functional limit under additional continuity conditions and verifies them for a logistic example.

3.2 Replacement decisions and deployed thresholds

The probability of a correct replacement decision depends on both the estimated machine frontier and the doctor posterior. We first relate these two errors to the original fixed-threshold rule, then bound the performance of the threshold actually chosen.

Let $\mathcal{C}_n \subseteq [0, 1]$ be the candidate set, containing $c = 1$. It is either a finite grid or the full threshold interval. In this subsection, $\widehat{g}_{\varepsilon}$ denotes the frontier over the candidates actually searched, and

$$\widehat{c}_j \in \arg \max_{c \in \mathcal{C}_n} \widehat{q}_j^{\varepsilon}(c), \quad \widehat{q}_{\max, j}^{\varepsilon} = \widehat{q}_j^{\varepsilon}(\widehat{c}_j). \quad (3.4)$$

We use a measurable choice of maximizer. A finite grid with a fixed tie-breaking rule meets this requirement. On the full interval, posterior maximization additionally requires attainment, as in Definition 3; continuity of both robust components on the compact search interval is one sufficient condition under (2.17). Lemma 2 gives the equivalent frontier representation. A grid has its own frontier: its approximation error relative to the population continuous-threshold frontier is included below.

Assumption 2 (Separation for replacement). *The number of doctors J is fixed. There are $\delta_0 > 0$ and $d_H > 0$ such that $\alpha_{H, j} \in [\alpha_l + d_H, \alpha_u - d_H]$ for every j , and either*

$$\beta_{H, j} \leq g_{\varepsilon}(\alpha_{H, j}; P_0) - \delta_0 \quad \text{or} \quad \beta_{H, j} \geq g_{\varepsilon}(\alpha_{H, j}; P_0) + \delta_0.$$

This condition separates a finite set of doctors from the frontier. It is distinct from a distributional margin condition over a population of doctors.

Assumption 3 (Regularity of the frontier in FPR index). *On $[\alpha_l, \alpha_u] \subset (0, 1)$, the frontier $\alpha \mapsto g_{\varepsilon}(\alpha; P_0)$ is Lipschitz with constant $L_g \geq 0$.*

Define the random posterior tail

$$T_j(t) := \Pi_j(\|\theta_H - \theta_{H,j}\|_\infty > t \mid \mathcal{D}_j). \quad (3.5)$$

Its center is the true human pair, whereas the posterior itself varies with the historical data.

Assumption 4 (Mean posterior concentration). *For each doctor with historical sample size $n_j \geq 1$, there are constants $C_j \geq 1$ and $c_j > 0$, independent of n_j and t , such that*

$$\mathbb{E}_{P_0}[T_j(t)] \leq C_j \exp(-c_j n_j t^2), \quad 0 < t \leq 1. \quad (3.6)$$

The expectation is over repeated reference-environment histories $\mathcal{D}_j$, conditional on the fixed doctor type. Write $C_0 = \max_j C_j$, $c_0 = \min_j c_j$, and $n_{\min} = \min_j n_j$.

The expectation in (3.6) allows an unusual history to produce a posterior far from the truth. Lemma 6 verifies the condition for independent historical cases, prevalence bounded away from zero and one, and the fixed Dirichlet prior in (2.17). It does not require independence between doctors for the subsequent union bounds.

Theorem 2 (Consistency of the fixed-threshold replacement rule). *Maintain the human comparison in (2.4), the implementation in (3.4), and Assumptions 2–4. Let*

$$r_n := \sup_{\alpha \in [\alpha_l, \alpha_u]} |\hat{g}_\varepsilon(\alpha) - g_\varepsilon(\alpha; P_0)|, \quad 0 < t_0 \leq \min \left\{ \frac{d_H}{2}, \frac{\delta_0}{4(1 + L_g)} \right\}.$$

On $\{r_n \leq \delta_0/4\}$:

(i) *if $\beta_{H,j} \leq g_\varepsilon(\alpha_{H,j}; P_0) - \delta_0$, then $\hat{q}_{\max,j}^\varepsilon \geq 1 - T_j(t_0)$;*

(ii) *if $\beta_{H,j} \geq g_\varepsilon(\alpha_{H,j}; P_0) + \delta_0$, then $\hat{q}_{\max,j}^\varepsilon \leq T_j(t_0)$.*

Thus $\hat{q}_{\max,j}^\varepsilon \rightarrow_p 1$ or 0, respectively, when $r_n \rightarrow_p 0$ and $n_{\min} \rightarrow \infty$. For $\bar{\tau} = \min\{\tau, 1 - \tau\}$, the probability that at least one doctor is assigned the wrong replacement decision obeys

$$\mathbb{P}(\text{at least one wrong decision}) \leq \mathbb{P}(r_n > \delta_0/4) + \frac{JC_0}{\bar{\tau}} \exp(-c_0 n_{\min} t_0^2). \quad (3.7)$$

Here “wrong” refers to the population robust-dominance classification in Assumption 2.

The first term in (3.7) is machine-frontier error; the second is doctor-estimation error. The proof chooses one rectangle containing a posterior neighborhood for a dominated doctor. For a non-dominated doctor, every candidate rectangle misses that neighborhood. The required convergence of r_n includes any grid or component approximation used in implementation. A fixed coarse grid need not approximate the continuous frontier as the sample grows.

Theorem 3 (Performance of the selected deployment threshold). *Maintain the human comparison in (2.4). Let $\hat{c}_j$ be an actually selected candidate satisfying $\hat{q}_j^{\varepsilon}(\hat{c}_j) \geq \tau$, and define*

$$\begin{aligned} e_{\alpha,n} &:= \sup_{c \in \mathcal{C}_n} \left| \hat{\alpha}_{\varepsilon}(c) - \bar{\alpha}_{\varepsilon}(c; P_0) \right|, \\ e_{\beta,n} &:= \sup_{c \in \mathcal{C}_n} \left| \hat{\beta}_{\varepsilon}(c) - \underline{\beta}_{\varepsilon}(c; P_0) \right|. \end{aligned} \quad (3.8)$$

For every $Q \in \mathcal{B}_{\varepsilon}(P_0)$,

$$\begin{aligned} \text{FPR}_Q(\hat{c}_j) &\leq \hat{\alpha}_{\varepsilon}(\hat{c}_j) + e_{\alpha,n}, \\ \text{TPR}_Q(\hat{c}_j) &\geq \hat{\beta}_{\varepsilon}(\hat{c}_j) - e_{\beta,n}. \end{aligned} \quad (3.9)$$

Consequently, under the doctor posterior, the event

$$\text{FPR}_Q(\hat{c}_j) \leq \alpha_H + e_{\alpha,n}, \quad \text{TPR}_Q(\hat{c}_j) \geq \beta_H - e_{\beta,n} \quad \text{for every } Q \in \mathcal{B}_{\varepsilon}(P_0)$$

has probability at least τ .

For any $t > 0$ with $T_j(t) < \tau$, the true human pair satisfies

$$\text{FPR}_Q(\hat{c}_j) \leq \alpha_{H,j} + e_{\alpha,n} + t, \quad \text{TPR}_Q(\hat{c}_j) \geq \beta_{H,j} - e_{\beta,n} - t \quad (3.10)$$

uniformly over the same ambiguity set. Under Assumption 4, for $\delta \in (0, 1)$ put

$$t_n(\delta) := \min \left\{ 1, \sqrt{\frac{\log(2JC_0/(\delta\tau))}{c_0 n_{\min}}} \right\}. \quad (3.11)$$

With sampling probability at least $1 - \delta$, (3.10) holds with $t = t_n(\delta)$ for every replaced doctor and every $Q \in \mathcal{B}_{\varepsilon}(P_0)$.

Theorem 3 evaluates the selected threshold. Its posterior statement allows for component estimation error; the sampling statement also uses concentration of the human posterior.

3.3 Welfare and regret

We compare the implemented rule with retaining all doctors and with the deployment matched-FPR benchmark. The first comparison concerns realized harm. The second also includes forgone gains and differences between the thresholds used by the two policies. Correctly choosing the same replacement set does not remove those threshold differences.

For fixed weights $\phi, \eta \geq 0$, we measure predictive welfare by the value of true positives less the cost of false positives. A per-case payoff interpretation incorporates class prevalence

in these weights. Define

$$W_j(c; Q) := \phi \text{TPR}_Q(c) - \eta \text{FPR}_Q(c), \quad W_{H,j}(Q) := \phi \beta_{H,j} - \eta \alpha_{H,j}.$$

If a set of doctors $\mathcal{R}$ is replaced using a threshold profile $c = (c_j)_{j \in \mathcal{R}}$, aggregate welfare is

$$W(\mathcal{R}, c; Q) := \frac{1}{J} \sum_{j \in \mathcal{R}} W_j(c_j; Q) + \frac{1}{J} \sum_{j \notin \mathcal{R}} W_{H,j}(Q). \quad (3.12)$$

For the deployment law Q , define the matched-FPR dominance benchmark set

$$\mathcal{R}^*(Q) := \{j : \beta_{H,j} < g_Q(\alpha_{H,j})\},$$

the population robust replacement set

$$\mathcal{R}_\varepsilon(P_0) := \{j : \beta_{H,j} < g_\varepsilon(\alpha_{H,j}; P_0)\},$$

and the sample robust replacement set

$$\mathcal{R}_\varepsilon^{(n)} := \{j : \hat{q}_{\max,j}^\varepsilon \geq \tau\}.$$

Let $\hat{c} = (\hat{c}_j)_{j \in \mathcal{R}_\varepsilon^{(n)}}$ denote the threshold profile selected by the sample robust rule, where each $\hat{c}_j$ is the candidate selected by (3.4).

The matched-FPR welfare comparison is defined at deployment laws $Q \in \mathcal{B}_\varepsilon(P_0)$ for which the following thresholds exist. For each $j \in \mathcal{R}^*(Q)$, let $c_j^*(Q)$ denote a threshold on the deployment ROC satisfying $\text{FPR}_Q(c_j^*(Q)) = \alpha_{H,j}$ and $\text{TPR}_Q(c_j^*(Q)) = g_Q(\alpha_{H,j})$, and write $c^*(Q) := (c_j^*(Q))_{j \in \mathcal{R}^*(Q)}$. Uniform statements about this matched-FPR comparison below concern these laws. The all-human comparison does not require matched-FPR attainment.

The next condition sharpens the bound for differences in the replacement set. For a fixed group of doctors it imposes separation uniformly over deployment environments.

Assumption 5 (Margin condition for the finite doctor collection). *There exist constants $C_M, \gamma_M > 0$ such that, for every $Q \in \mathcal{B}_\varepsilon(P_0)$ and every $t > 0$,*

$$\frac{1}{J} \sum_{j=1}^J \mathbb{I}(|g_Q(\alpha_{H,j}) - \beta_{H,j}| \leq t) \leq C_M t^{\gamma_M}. \quad (3.13)$$

Because J is fixed, this condition requires uniform separation across the entire deployment set:

$$\inf_{Q \in \mathcal{B}_\varepsilon(P_0)} \min_{j \leq J} |g_Q(\alpha_{H,j}) - \beta_{H,j}| \geq (JC_M)^{-1/\gamma_M} > 0.$$

Indeed, a single doctor at distance $d > 0$ contributes $1/J$ when $t = d$. A zero distance

violates the condition as $t \downarrow 0$. Thus the assumption excludes a deployment path that moves a human point through the frontier. It is used only to sharpen the replacement-set terms; Theorem 4 does not require it. The constants may depend on the fixed group and radius.

Theorem 4 (Welfare relative to retaining all doctors). *Maintain the human comparison in (2.4) and Assumption 4. Replace exactly when $\mathcal{C}_j^\tau$ in (2.15) is nonempty and use a measurable threshold in that set. Let $\phi, \eta \geq 0$ and $\delta \in (0, 1)$. With sampling probability at least $1 - \delta$, simultaneously for every $Q \in \mathcal{B}_\varepsilon(P_0)$,*

$$W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q) - W(\emptyset, \emptyset; Q) \geq -\frac{|\mathcal{R}_\varepsilon^{(n)}|}{J} [\phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi + \eta)t_n(\delta)], \quad (3.14)$$

where the errors and $t_n(\delta)$ are defined in Theorem 3. The bound covers every actually replaced doctor. It uses posterior concentration and component accuracy.

For the matched-FPR comparison, partition the replacement decisions into common, missed, and extra replacements:

$$\mathcal{I}_n = \mathcal{R}^*(Q) \cap \mathcal{R}_\varepsilon^{(n)}, \quad \mathcal{M}_n = \mathcal{R}^*(Q) \setminus \mathcal{R}_\varepsilon^{(n)}, \quad \mathcal{A}_n = \mathcal{R}_\varepsilon^{(n)} \setminus \mathcal{R}^*(Q).$$

Then the complete comparison is

$$\begin{aligned} & W(\mathcal{R}^*, c^*; Q) - W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q) \\ &= \frac{1}{J} \sum_{j \in \mathcal{I}_n} [W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q)] \\ & \quad + \frac{1}{J} \sum_{j \in \mathcal{M}_n} [W_j(c_j^*; Q) - W_{H,j}(Q)] \\ & \quad + \frac{1}{J} \sum_{j \in \mathcal{A}_n} [W_{H,j}(Q) - W_j(\widehat{c}_j; Q)]. \end{aligned} \quad (3.15)$$

The first sum remains when the replacement sets coincide. Its summands are signed: the matched-FPR benchmark need not maximize $\phi \text{TPR} - \eta \text{FPR}$ over unrestricted thresholds.

Definition 4 (Matched-FPR welfare comparison). *For the implemented posterior rule, define the signed welfare difference*

$$\text{Regret}(\varepsilon, Q) := W(\mathcal{R}^*(Q), c^*(Q); Q) - W(\mathcal{R}_\varepsilon^{(n)}, \widehat{c}; Q).$$

For a separate population comparison, suppose that each $j \in \mathcal{R}_\varepsilon(P_0)$ has a threshold $c_{\varepsilon,j}$ with

$$\overline{\alpha}_\varepsilon(c_{\varepsilon,j}; P_0) = \alpha_{H,j}, \quad \underline{\beta}_\varepsilon(c_{\varepsilon,j}; P_0) = g_\varepsilon(\alpha_{H,j}; P_0).$$

Write $c_\varepsilon = (c_{\varepsilon,j})_{j \in \mathcal{R}_\varepsilon(P_0)}$ and define

$$\text{Regret}_{\text{pop}}(\varepsilon, Q) := W(\mathcal{R}^*(Q), c^*(Q); Q) - W(\mathcal{R}_\varepsilon(P_0), c_\varepsilon; Q). \quad (3.16)$$

The population comparison matches the robust FPR to the true human FPR. The implemented rule instead selects its threshold from the human posterior. Both are compared with the deployment matched-FPR benchmark.

The threshold contribution can be bounded using quantities recorded when the decision is made. Let $a_{n,j}$ be a median of doctor j 's FPR posterior. For the Dirichlet model this is the unique median of the Beta distribution in (2.17). Define

$$\begin{aligned} \hat{K}_n := \frac{1}{J} \sum_{j \in \mathcal{R}_\varepsilon^{(n)}} & \left[\phi\{\hat{g}_\varepsilon(a_{n,j}) - \hat{\beta}_\varepsilon(\hat{c}_j)\} \right. \\ & \left. + \eta\{\hat{\alpha}_\varepsilon(\hat{c}_j) - a_{n,j}\} \right]_+, \quad [x]_+ := \max\{x, 0\}. \end{aligned} \quad (3.17)$$

This quantity compares the estimated frontier value at the posterior median FPR with the selected threshold's value. The median is a reference point for the welfare bound, and the empirical FPR constraint may be slack there.

The two threshold choices in (2.15) have the same replacement set and the same posterior medians. Since

$$\hat{K}_n(c) = \frac{1}{J} \sum_{j \in \mathcal{R}_\varepsilon^{(n)}} [\phi\hat{g}_\varepsilon(a_{n,j}) - \eta a_{n,j} - \hat{V}_\varepsilon(c_j)]_+,$$

(2.16) implies

$$\hat{K}_n(\hat{c}^W) \leq \hat{K}_n(\hat{c}^a). \quad (3.18)$$

The bounds for the selected threshold, all-human welfare and the sample matched-FPR comparison also hold for $\hat{c}^W$: their proofs use posterior feasibility of the deployed threshold and replacement whenever a feasible candidate exists. The replacement indicators in Theorem 2 are identical. Its convergence statement for the maximum posterior probability continues to concern that maximum, rather than the posterior probability of a different feasible threshold. Equation (3.18) compares terms in the welfare bound; actual deployment welfare is evaluated separately.

Theorem 5 (Welfare bounds with the selected threshold retained). *Maintain (2.4), the actual rule (3.4), and the matched-FPR attainment stipulated above. The candidate set is either the full threshold interval or a finite set containing the all-negative rule, and the stated posterior maximum is attained. Let $\phi, \eta \geq 0$ and suppose Assumption 5 holds. Set*

$$\Delta_\varepsilon(Q) := \sup_{\alpha \in [\alpha_l, \alpha_u]} \{g_Q(\alpha) - g_\varepsilon(\alpha; P_0)\} \geq 0.$$

All human FPRs lie in $[\alpha_l + d_H, \alpha_u - d_H]$ for a fixed $d_H > 0$.

(a) If the population robust matched-FPR thresholds in Definition 4 exist, then, for every $Q \in \mathcal{B}_\varepsilon(P_0)$ at which the matched-FPR benchmark is defined,

$$\text{Regret}_{\text{pop}}(\varepsilon, Q) \leq \phi \frac{|\mathcal{R}_\varepsilon(P_0)|}{J} \Delta_\varepsilon(Q) + \phi C_M \Delta_\varepsilon(Q)^{1+\gamma_M}. \quad (3.19)$$

(b) Suppose also that Assumptions 3 and 4 hold. For $\delta \in (0, 1)$, let $\bar{\tau} := \min\{\tau, 1 - \tau\}$ and

$$t := \sqrt{\frac{\log\{2JC_0/(\delta\bar{\tau})\}}{c_0 n_{\min}}} \leq \min\{1, d_H\}. \quad (3.20)$$

Let r_n be the frontier error in Theorem 2, including any candidate-grid approximation, and let $e_{\alpha,n}, e_{\beta,n}$ be the component errors in Theorem 3. Define the nonnegative tolerances

$$\begin{aligned} h_n &:= r_n + (1 + L_g)t, \\ u_n &:= \phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi + \eta)t, \\ v_n &:= \phi r_n + \phi e_{\beta,n} + \eta e_{\alpha,n} + (\phi L_g + \eta)t, \end{aligned}$$

and the replacement fraction $\hat{\pi}_n := |\mathcal{R}_\varepsilon^{(n)}|/J$. With sampling probability at least $1 - \delta$, simultaneously for every $Q \in \mathcal{B}_\varepsilon(P_0)$ at which the matched-FPR benchmark is defined,

$$\begin{aligned} \text{Regret}(\varepsilon, Q) &\leq \hat{K}_n + \hat{\pi}_n \{\phi \Delta_\varepsilon(Q) + v_n\} \\ &\quad + \phi C_M \{\Delta_\varepsilon(Q) + h_n\}^{1+\gamma_M} + C_M u_n h_n^{\gamma_M}. \end{aligned} \quad (3.21)$$

The statement also holds for the welfare choice $\hat{c}^W$ in (2.15), with its own $\hat{K}_n$ and the same replacement set. On the same event, the fractions of extra and missed replacements obey

$$\frac{|\mathcal{A}_n|}{J} \leq C_M h_n^{\gamma_M}, \quad \frac{|\mathcal{M}_n|}{J} \leq C_M \{\Delta_\varepsilon(Q) + h_n\}^{\gamma_M}. \quad (3.22)$$

The probability is over the observed histories and validation data, conditional on the fitted score. The component and frontier errors in the bound are random theoretical quantities. No independence between those errors and the doctor posteriors is required. A quantitative bound for them is needed to turn (3.21) into a numerical confidence guarantee.

The last two terms of (3.21) bound differences in the replacement set. The first two retain threshold differences among doctors replaced by both rules. Increasing the sample size reduces estimation error, but need not make the threshold maximizing posterior

dominance coincide with the matched-FPR threshold. Consequently, $\widehat{K}_n$ need not vanish. The following example isolates this distinction within the smooth-score, positive-radius model.

Example 1 (Safety and matched-FPR thresholds). *Let $\mathcal{X} = \mathbb{R}$, $P_0(Y = 1) = 1/2$, and $X \mid Y = y \sim \text{Unif}[0, 1]$ for both classes. Use $m(x) = (1 + e^{-x})^{-1}$, one doctor with stable pair $(\alpha_H, \beta_H) = (0.8, 0.2)$, and $(\phi, \eta) = (2, 1)$. Write $s = 2\sqrt{\varepsilon} \in (0, 0.1)$ and let $u = 1 - \log\{c/(1 - c)\}$ denote the nominal rate at thresholds with $\log\{c/(1 - c)\} \in [0, 1]$. For $s \leq u \leq 1 - s$,*

$$\bar{\alpha}_\varepsilon(c; P_0) = u + s, \quad \underline{\beta}_\varepsilon(c; P_0) = u - s.$$

At deployment $Q = P_0$, the matched-FPR benchmark uses $u = 0.8$, whereas the population robust matched-FPR comparison uses $u = 0.8 - s$. Both replace the doctor, yet

$$\text{Regret}_{\text{pop}}(\varepsilon, P_0) = s.$$

On $[0.7, 0.9]$, the frontier gap is $\Delta_\varepsilon(P_0) = 2s$, and the finite-doctor margin condition holds uniformly over the ambiguity set with $\gamma_M = 1$ and $C_M = (0.6 - 2s)^{-1}$.

Now suppose the reference machine frontier is known, while the doctor's iid history grows and is analyzed with the fixed positive Dirichlet prior. Any exact maximizer of the fixed-threshold posterior probability has nominal rate $\widehat{u} \rightarrow 0.5$ almost surely, and replacement occurs eventually for every fixed $\tau < 1$. Thus the implemented rule has

$$\text{Regret}(\varepsilon, P_0) \longrightarrow 0.3, \quad \widehat{K}_n \longrightarrow 0.3 - s.$$

The population comparison differs at order Δ_ε , while the posterior-maximizing rule has a persistent threshold difference. In this symmetric example, the latter threshold balances the two posterior tail decay rates, independently of the welfare weights. Appendix A.8 proves both statements.

Appendix A.7 records the resulting stochastic order for the one-sided remainder. The bound retains both the threshold contribution and the deployment-frontier difference; fixed-radius accuracy alone does not make the full welfare difference vanish.

4 An extension to local input perturbations

4.1 A bound on each observation's displacement

The Wasserstein model bounds average covariate displacement. This extension bounds each observation's displacement, a restriction related to the input-perturbation approach of Madry et al. (2017). It also provides a way to evaluate local changes for random forests

(RF) and other tree scores. We use ρ for the per-observation limit and ε for the average budget.

Definition 5 (Pointwise-budget ambiguity). *Let $\mathcal{X}$ be a closed subset of $\mathbb{R}^d$. For $\rho \geq 0$, write $K_\rho(x) = \{z \in \mathcal{X} : \|z - x\|_2 \leq \rho\}$ and define*

$$\mathcal{B}_\rho^{\text{pt}}(P) := \{(T_X, \text{Id}_Y)_\# P : T_X \text{ is Borel measurable and } T_X(X, Y) \in K_\rho(X) \text{ } P\text{-a.s.}\}.$$

The pushforward is the law of $(T_X(X, Y), Y)$. Thus this definition retains deterministic, label-preserving perturbations, with a separate displacement limit for each observation.

Definition 6 (Worst-case score under direct perturbation). *For a fixed score $m : \mathcal{X} \rightarrow [0, 1]$, define*

$$\underline{m}_\rho(x) := \inf_{z \in K_\rho(x)} m(z), \quad (4.1)$$

$$\overline{m}_\rho(x) := \sup_{z \in K_\rho(x)} m(z). \quad (4.2)$$

We use these scores when they are measurable and their extrema admit measurable attaining selections $z_\rho^-(x), z_\rho^+(x) \in K_\rho(x)$. This holds for continuous m on closed $\mathcal{X}$, and for a fixed finite ensemble of axis-aligned trees on $\mathbb{R}^d$ with a specified convention at split boundaries; the proof below verifies both cases.

Definition 7 (Population and empirical direct perturbation frontier). *For $0 < p = P(Y = 1) < 1$, define the strict-threshold components*

$$\tilde{\beta}_\rho(c; P) = P(\underline{m}_\rho(X) > c \mid Y = 1), \quad \tilde{\alpha}_\rho(c; P) = P(\overline{m}_\rho(X) > c \mid Y = 0),$$

and the deterministic-threshold frontier

$$\tilde{g}_\rho(\alpha; P) = \sup_{c \in [0, 1]} \{\tilde{\beta}_\rho(c; P) : \tilde{\alpha}_\rho(c; P) \leq \alpha\}, \quad \alpha \in [0, 1]. \quad (4.3)$$

Let $N_y = \sum_{i=1}^n \mathbb{I}(Y_i = y)$. On samples with both classes, its empirical counterpart is

$$\tilde{g}_\rho(\alpha; \hat{\mathbb{P}}_n) = \sup_{c \in [0, 1]} \left\{ \frac{1}{N_1} \sum_{i: Y_i=1} \mathbb{I}(\underline{m}_\rho(X_i) > c) : \frac{1}{N_0} \sum_{i: Y_i=0} \mathbb{I}(\overline{m}_\rho(X_i) > c) \leq \alpha \right\}. \quad (4.4)$$

The extrema and the score are fixed when estimating this frontier. The threshold $c = 1$ is always feasible. No interpolation between different thresholds is used.

Proposition 3 (Pointwise probabilities and frontier ordering). *Suppose the measurable attaining selections in Definition 6 exist, P has a finite first moment, and $0 < P(Y =$*

1) < 1 . Then, for every c ,

$$\inf_{Q \in \mathcal{B}_\rho^{\text{pt}}(P)} \text{TPR}_Q(c) = \tilde{\beta}_\rho(c; P), \quad (4.5)$$

$$\sup_{Q \in \mathcal{B}_\rho^{\text{pt}}(P)} \text{FPR}_Q(c) = \tilde{\alpha}_\rho(c; P). \quad (4.6)$$

Moreover, $\tilde{g}_0(\alpha; P) = g(\alpha; P)$, and for $0 \leq \rho \leq \varepsilon$,

$$g_\varepsilon(\alpha; P) \leq \tilde{g}_\rho(\alpha; P) \leq g(\alpha; P). \quad (4.7)$$

The ordering follows from set inclusion: the average budget permits concentrated displacement that the pointwise limit excludes. Measurable attainment ensures that the strict-threshold probabilities represent attainable classifications; Appendix A.9 gives a counterexample when attainment fails.

Exact local perturbations for a logistic score. For the logistic score on $\mathbb{R}^d$ in (2.19), write $\sigma(t) = (1 + e^{-t})^{-1}$. Cauchy–Schwarz and monotonicity give

$$\underline{m}_\rho(x) = \sigma\{z(x) - \rho\|\beta\|_2\}, \quad \overline{m}_\rho(x) = \sigma\{z(x) + \rho\|\beta\|_2\}. \quad (4.8)$$

Using the index distributions in Remark 1,

$$\tilde{\alpha}_\rho(c) = 1 - H_0(a_c - \rho), \quad \tilde{\beta}_\rho(c) = 1 - H_1(a_c + \rho). \quad (4.9)$$

When H_0 is continuous and strictly increasing at the relevant quantiles, set $a_\alpha = H_0^{-1}(1 - \alpha)$. The FPR constraint binds at $a_c = a_\alpha + \rho$, giving

$$\tilde{g}_\rho(\alpha) = 1 - H_1(a_\alpha + 2\rho), \quad g(\alpha) - \tilde{g}_\rho(\alpha) = 2H'_1(a_\alpha)\rho + o(\rho). \quad (4.10)$$

The expansion uses continuity of H'_1 at a_α . Unlike the average budget’s square-root effect, this local effect is linear in ρ .

One map attains both component extremes, at every threshold:

$$T_\rho(x, y) = x + (1 - 2y)\rho \frac{\beta}{\|\beta\|_2}. \quad (4.11)$$

It moves positive cases against, and negative cases along, the fitted index. Consequently, for $\phi, \eta \geq 0$,

$$\inf_{Q \in \mathcal{B}_\rho^{\text{pt}}(P)} \{\phi \text{TPR}_Q(c) - \eta \text{FPR}_Q(c)\} = \phi \tilde{\beta}_\rho(c; P) - \eta \tilde{\alpha}_\rho(c; P). \quad (4.12)$$

The identity also holds with P replaced by the empirical distribution, giving the welfare-

based threshold its worst-case interpretation. At the same numerical radius, $\mathcal{B}_\rho^{\text{pt}}(P) \subseteq \mathcal{B}_\rho(P)$, where $\mathcal{B}_\rho(P)$ denotes the Wasserstein set in (2.7) evaluated at $\varepsilon = \rho$. The pointwise calculation rules out the concentration of transport allowed by the average budget.

In computation, a finite set of admissible proposals that includes the original observation gives scores $\underline{m}_{\rho,K}, \overline{m}_{\rho,K}$ satisfying

$$\underline{m}_\rho \leq \underline{m}_{\rho,K} \leq m \leq \overline{m}_{\rho,K} \leq \overline{m}_\rho.$$

For the same observations and thresholds, this approximation lies between the exact pointwise and nominal frontiers. The finite search is optimistic: a larger validation sample alone does not remove its optimization error.

For tree scores on $\mathcal{X} = \mathbb{R}^d$, we evaluate the original input and K proposals per observation. Each proposal adds a uniformly drawn unit direction multiplied by an independent $\text{Unif}[0, \rho]$ radius. The sampling is uniform in radius, rather than in ball volume. The observed extrema enter the posterior calculation, and the selected threshold is then deployed. Section 4.2 analyzes the exact extrema.

4.2 Estimation with continuous and discrete scores

With the pointwise radius ρ and fitted score fixed, define

$$Z_0(x) = \overline{m}_\rho(x), \quad Z_1(x) = \underline{m}_\rho(x), \quad F_y(c) = P_0(Z_y(X) \leq c \mid Y = y).$$

Then $\tilde{\alpha}_\rho = 1 - F_0$ and $\tilde{\beta}_\rho = 1 - F_1$. Let $\hat{F}_y$ be the corresponding class-conditional empirical CDF. These two one-dimensional distributions determine the extension. Their estimation is distinct from computing the local score extrema.

Assumption 6 (Sampling and local regularity for the pointwise frontier). *The validation sample is iid from P_0 , independent of the training sample on which we condition, with $0 < p_0 = P_0(Y = 1) < 1$. The fixed exact extrema in Definition 6 are measurable. For a continuous-frontier result on $I = [\alpha_l, \alpha_u] \subset (0, 1)$, assume there is $C = [c_l, c_u] \subset (0, 1)$ and $b > 0$ such that*

$$1 - F_0(c_l) > \alpha_u + b, \quad 1 - F_0(c_u) < \alpha_l - b,$$

F_0, F_1 are continuously differentiable on a neighborhood of C , and

$$0 < \kappa \leq f_0(c), \quad 0 \leq f_1(c) \leq L_1 < \infty \quad (c \in C),$$

where $f_y = F'_y$. The differentiability and density conditions are used only for the continuous-frontier conclusion, not for component estimation or the finite-support conclusion below.

Theorem 6 (Statistical theory for the exact pointwise frontier). *Under the sampling conditions in Assumption 6, write $d_{y,n} = \|\widehat{F}_y - F_y\|_\infty$ and $\widehat{g}_\rho = \tilde{g}_\rho(\cdot; \widehat{\mathbb{P}}_n)$.*

(i) *For continuous or discrete score distributions, $d_{0,n} + d_{1,n} = O_p(n^{-1/2})$. On samples with both classes, whenever $\alpha \pm d_{0,n} \in [0, 1]$,*

$$\tilde{g}_\rho(\alpha - d_{0,n}; P_0) - d_{1,n} \leq \widehat{g}_\rho(\alpha) \leq \tilde{g}_\rho(\alpha + d_{0,n}; P_0) + d_{1,n}. \quad (4.13)$$

(ii) *Under the additional density and interval conditions,*

$$\sup_{\alpha \in I} |\widehat{g}_\rho(\alpha) - \tilde{g}_\rho(\alpha; P_0)| = O_p(n^{-1/2}).$$

As a function of (F_0, F_1) , the frontier map into $\ell^\infty(I)$ is Hadamard differentiable in the uniform norm on C , tangentially to continuous CDF perturbation directions. With $c_\alpha = F_0^{-1}(1 - \alpha)$, its derivative is

$$\dot{\tilde{g}}_{(h_0, h_1)}(\alpha) = -h_1(c_\alpha) + \frac{f_1(c_\alpha)}{f_0(c_\alpha)} h_0(c_\alpha). \quad (4.14)$$

(iii) *Suppose instead that the two score laws have fixed finite supports. Let $s_1 < \dots < s_K$ list their union, let $c_1 < \dots < c_M$ list the distinct elements of $\{0, s_1, \dots, s_K, 1\}$, and set*

$$A_k = 1 - F_0(c_k), \quad B_k = 1 - F_1(c_k), \quad \mathcal{A} = \{A_k : 1 \leq k \leq M\}.$$

For any compact $I \subset (0, 1)$ with $\text{dist}(I, \mathcal{A}) > 0$, the same uniform root- n frontier rate holds. Relative to perturbations of the two conditional mass vectors on these fixed supports, the map into $\ell^\infty(I)$ is Hadamard differentiable. Writing $v_y = (v_{y,1}, \dots, v_{y,K})$ for a mass-vector direction and $k(\alpha) = \min\{k : A_k \leq \alpha\}$, its derivative is

$$\dot{\tilde{g}}_{(v_0, v_1)}(\alpha) = \sum_{\ell: s_\ell > c_{k(\alpha)}} v_{1,\ell}. \quad (4.15)$$

At a binding FPR level in $\mathcal{A}$, even pointwise consistency of the frontier can fail. Consequently, finite support alone does not give uniform frontier convergence or directional differentiability on every interior FPR interval.

For a fixed finite tree ensemble, component accuracy transfers to frontier accuracy on FPR regions separated from the finite feasibility boundaries. For a continuous score, the density conditions give the inverse's local stability. Finite perturbation search contributes an additional optimization error to these statistical errors.

Corollary 1 (Replacement and welfare under exact pointwise perturbations). *Maintain the stable-human comparison (2.4) and the measurable attaining extrema of Proposition 3.*

Fix $\rho \geq 0$. Use the same posterior and a candidate set containing the all-negative rule, with attained frontiers and measurable posterior maximizers. Replace the Wasserstein objects by

$$(\mathcal{B}_\varepsilon, g_\varepsilon, \overline{\alpha}_\varepsilon, \underline{\beta}_\varepsilon) \mapsto (\mathcal{B}_\rho^{\text{pt}}, \tilde{g}_\rho, \tilde{\alpha}_\rho, \tilde{\beta}_\rho).$$

Define $\tilde{r}_n$, $\tilde{e}_{\alpha,n}$, $\tilde{e}_{\beta,n}$, $\tilde{\Delta}_\rho(Q)$ and $\hat{K}_n$ by the same substitution in (3.4), Theorems 2–5 and (3.17). In particular, the frontier error includes the candidate-grid approximation. Then:

- (i) Under Assumptions 2–4, with separation and Lipschitz regularity now imposed on $\tilde{g}_\rho$, the posterior-maximum and replacement-consistency conclusions of Theorem 2 hold with error $\tilde{r}_n$.
- (ii) For a selected candidate whose pointwise posterior probability is at least τ , Theorem 3’s posterior performance statement holds over $\mathcal{B}_\rho^{\text{pt}}(P_0)$ with errors $\tilde{e}_{\alpha,n}, \tilde{e}_{\beta,n}$. Under Assumption 4, its sampling statement and Theorem 4’s all-human welfare bound hold with those errors and the same $t_n(\delta)$.
- (iii) Theorem 5 holds with the substituted objects when its interior conditions and matching thresholds hold, Assumption 5 is imposed over $Q \in \mathcal{B}_\rho^{\text{pt}}(P_0)$, and, for part (b), Assumptions 3 and 4 hold for that frontier. The sample welfare conclusions also apply to the feasible welfare choice in (2.15).

Proof. Proposition 3 gives exact worst-case components, so the substituted version of (3.9) holds for every admitted Q . The minimum feasible threshold attains the pointwise frontier by the right-continuity of its FPR component and monotonicity of both components. Every feasible robust threshold is feasible for the deployment FPR constraint and has deployment TPR at least its lower component; hence $\tilde{g}_\rho \leq g_Q$. The fixed-rectangle argument, posterior concentration event and welfare decomposition in Appendices A.6 and A.7 then apply with the stated conditions and substitutions. Posterior feasibility and replacing whenever a feasible candidate exists are sufficient for the sample welfare steps. $\square$

For the logistic formula (4.10), the frontier derivative is

$$\tilde{g}'_\rho(\alpha) = \frac{H'_1(a_\alpha + 2\rho)}{H'_0(a_\alpha)}.$$

A positive lower bound on the denominator and a finite upper bound on the numerator over the stated interval supply its Lipschitz constant. With continuously differentiable H_0, H_1 on the corresponding shifted index neighborhoods and the stated interval and sampling conditions, these supply the continuous-score conditions of Theorem 6. Its rate concerns exact extrema; finite RF search additionally requires control of optimization

error. The inequalities for the posterior and empirical-reference quantities retain their separate sampling-error terms in the population welfare comparison.

5 Monte Carlo Simulations

5.1 A smooth-score experiment with known transport distance

The main experiment separates changes in the machine’s recorded predictors from the clinical information used by the doctor. In the reference environment, let $Z \sim N(0, I_5)$ and $U \sim N(0, \sigma_u^2)$ be independent, and set

$$R = b_0 + Z^\top \beta + U, \quad p(Z, U) = \{1 + \exp(-R)\}^{-1}, \quad Y \mid Z, U \sim \text{Bernoulli}(p(Z, U)). \quad (5.1)$$

The machine observes $X = Z$. Doctor j uses the clinical information (Z, U) and makes decision

$$\hat{Y}_j = \mathbb{I}\{p(Z, U) + \nu_j > c_{H,j}\}, \quad \nu_j \sim N(0, \sigma_\nu^2), \quad c_{H,j} = 0.15 + 0.7V_j, \quad V_j \sim \text{Beta}(2, 3).$$

The noises and doctor thresholds are independent of the reference cases. We retain $\beta = (1, 0.5, -0.3, 0.2, -0.1)^\top$, $b_0 = -1.5$, $\sigma_u = 0.15$ and $\sigma_\nu = 0.225$.

At deployment, only the recorded predictors change:

$$X_\delta = Z + (1 - 2Y)\delta e_1, \quad \delta \geq 0, \quad (5.2)$$

where e_1 is the first coordinate vector. The law of (Z, U, Y, ν_j) and each doctor’s decision remain unchanged. This construction represents class-dependent changes in a recorded field while clinical assessment uses stable information. It preserves prevalence and the human FPR/TPR pair; it does not impose invariance of $Y \mid X_\delta$.

Let Q_δ denote the resulting law of (X_δ, Y) . Conditional on each label, the reference feature law is translated by a vector of length δ . The translation coupling gives a conditional W_1 distance at most δ , while the norm of the conditional mean difference gives the reverse inequality. Consequently,

$$\mathcal{W}_\kappa(P_0, Q_\delta) = \delta. \quad (5.3)$$

Thus $Q_\delta \in \mathcal{B}_\varepsilon(P_0)$ precisely when $\delta \leq \varepsilon$. This statement concerns the population reference law; estimation of the frontier from validation data remains a separate source of error.

Each replication draws 10,000 training observations, 5,000 independent validation observations, and 500 historical observations for each of $J = 500$ doctors. We fit a logistic score using the training data alone and hold it fixed during validation and deployment.

Its ℓ_2 -penalty parameter is $C = 1$. The confusion-table prior uses $a_0 = 0.01$ per cell and the posterior cutoff is $\tau = 0.95$. For every $\varepsilon \in \{0, 0.05, 0.1, 0.2, 0.3\}$, we compute the exact empirical Wasserstein components on the same grid of 1,001 score thresholds from 0 to 1. Equation (2.18) selects a single candidate for each doctor, and deployment uses that same candidate. The radius-zero rule is the standard Bayesian comparison. Policies use this fixed threshold grid.

The deployment grid is $\delta \in \{0, 0.025, 0.05, 0.1, 0.3, 0.5, 1\}$; the two smallest positive values locate the interior and boundary of the main radius $\varepsilon = 0.05$. Within a replication, training data, validation data, doctor histories and all selected thresholds are held fixed across deployment distances. The experiment uses 100 independent replications and welfare weights $(\phi, \eta) \in \{(1, 1), (2, 1), (5, 1)\}$, with $(2, 1)$ as the main comparison. No welfare weight or deployment outcome enters the posterior threshold selection.

We evaluate conditional error rates by numerical integration under (5.1), rather than by an additional test sample. Appendix B gives the reduction to one-dimensional Gaussian integrals and the numerical checks. The resulting population rates also determine the matched-FPR benchmark at evaluation. For each policy we record its welfare relative to retaining all doctors, the replacement and harmful-replacement rates, and the common, missed and extra replacement terms in the signed welfare difference. The computable threshold term in Theorem 5 uses the recorded candidate frontier and the actual selected thresholds. Monte Carlo standard errors use replication-level observations, including paired differences between robust and standard rules. The experiment evaluates the deployment path in (5.2).

5.2 Replacement, welfare, and foregone gains

At every $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$, the rule retains all 500 doctors in all 100 replications. The standard rule replaces 41.51% on average, with a Monte Carlo standard error of 0.88 percentage points. At $\varepsilon = 0.05$, the largest posterior dominance probability across all doctors and replications is 5.85×10^{-10} , well below $\tau = 0.95$. No candidate comes close to satisfying the posterior requirement at this radius.

Retention avoids losses when the recorded-feature shift is large, but forgoes gains near the reference environment. In Table 1, the standard rule raises mean welfare from 0.6498 to 0.7323 at $\delta = 0$, a paired gain of 0.0826 (Monte Carlo standard error 0.0017). At the primary ambiguity-set boundary, $\delta = 0.05$, its gain is 0.0628 (0.0013). Mean standard welfare exceeds all-human welfare at every evaluated distance through 0.1, and is lower at 0.3, 0.5 and 1. At $\delta = 1$, the robust-minus-standard difference is 0.3226 (0.0068). In this grid, robust policies obtain that protection by retaining the doctors and forgoing the gains near the reference distribution.

Harm frequency and severity are distinct. Among doctors replaced by the standard

Table 1: Replacement welfare along the recorded-feature shift, $(\phi, \eta) = (2, 1)$.

δ	Standard welfare	Robust welfare	Robust minus standard	Harmful among standard-replaced (%)
0	0.7323 (0.0020)	0.6498 (0.0009)	-0.0826 (0.0017)	0.17 (0.03)
0.025	0.7225 (0.0019)	0.6498 (0.0009)	-0.0727 (0.0015)	0.31 (0.05)
0.05	0.7126 (0.0017)	0.6498 (0.0009)	-0.0628 (0.0013)	0.70 (0.08)
0.1	0.6926 (0.0014)	0.6498 (0.0009)	-0.0428 (0.0009)	2.82 (0.20)
0.3	0.6105 (0.0013)	0.6498 (0.0009)	0.0392 (0.0010)	99.94 (0.02)
0.5	0.5270 (0.0027)	0.6498 (0.0009)	0.1228 (0.0027)	100.00 (0.00)
1	0.3272 (0.0067)	0.6498 (0.0009)	0.3226 (0.0068)	100.00 (0.00)

Notes: Means over 100 independent replications; Monte Carlo standard errors are in parentheses. Welfare differences are paired within replications. Standard uses $\varepsilon = 0$; robust uses the primary $\varepsilon = 0.05$. The other prespecified positive radii, 0.1, 0.2 and 0.3, yield the same policy in every replication: no doctor is replaced. Robust welfare therefore equals all-human welfare. The harmful fraction among robust-replaced doctors is undefined, rather than zero. The laws with $\delta \leq 0.05$ lie in the primary population ambiguity set. Rates used in welfare evaluation are numerical population reference values from Appendix B.

rule, the mean harmful fraction is 0.17% at $\delta = 0$, 0.70% at 0.05, and 99.94% at 0.3. At both 0.5 and 1, every replaced doctor is harmed. Although that fraction is unchanged, the average loss per doctor across the full population rises from 0.1228 to 0.3226. The paired welfare differences have the same sign pattern under the two other prespecified welfare weights; Appendix B.1 reports them.

The matched-FPR comparison also distinguishes replacement decisions from threshold choices. At $\delta = 0$, the standard rule’s mean shortfall from that benchmark is 0.1718: 0.0286 comes from different thresholds among doctors replaced by both rules, and 0.1433 from missed replacements. The computable threshold term $\hat{K}_n$ averages 0.0327 (0.0008) for this rule; it is not the common-replacement welfare difference itself. For the positive-radius policies, $\hat{K}_n = 0$ because no doctor is replaced. Their benchmark shortfall at $\delta = 0.05$ is 0.1864, entirely from missed replacements.

5.3 Sensitivity to smaller radii

Smaller radii reveal the transition between the nominal policy and complete retention. After the grid in Table 1 yielded no replacements at positive radii, we examined the finer grid in Figure 2, holding the samples, decision rule and evaluation design fixed. Each radius defines a separate policy.

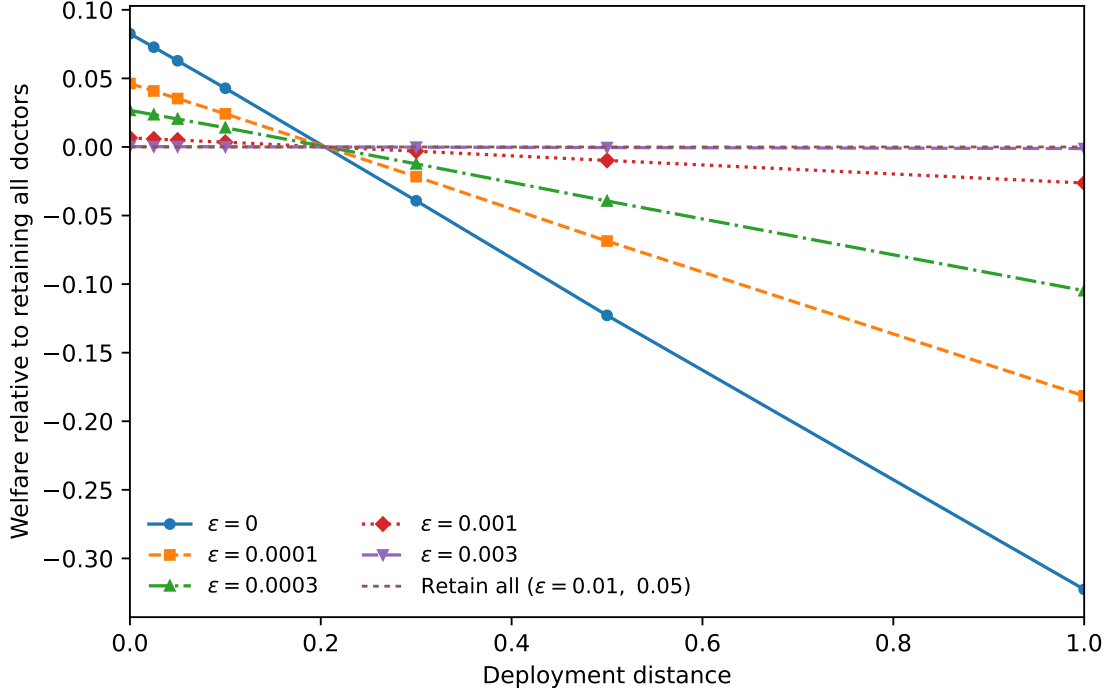


Figure 2: Replacement welfare across the evaluated deployment distances.
Notes: Each point is the mean of the policy’s welfare minus all-human welfare over the same 100 replications, with $(\phi, \eta) = (2, 1)$. The full smaller-radius range is shown; the policies at 0.01 and 0.05 coincide with retaining everyone. Lines connect evaluated distances. For positive radii below 0.025, only $\delta = 0$ is inside the corresponding ambiguity set among these evaluation points. Table 5 gives the full numerical entries and Monte Carlo standard errors.

At $\varepsilon = 0.0001$, the rule replaces 22.79% of doctors and gains 0.0463 relative to retaining everyone at $\delta = 0$, compared with the standard rule’s gain of 0.0826. At $\delta = 0.5$, its loss relative to all-human welfare is 0.0686 rather than the standard rule’s 0.1228. Increasing the radius further reduces both replacement and the magnitude of the welfare difference from retaining everyone. The radii 0.01 and 0.05 again give complete retention.

Across all three prespecified welfare weights, every positive radius with replacements has lower mean welfare than standard replacement at the evaluated distances through 0.1, but higher welfare than retaining all doctors. At 0.3, 0.5 and 1, the ranking reverses: partial replacement limits the standard policy’s loss but remains below all-human welfare. At both 0.5 and 1, every remaining replacement is harmful in each replication with replacements.

5.4 Two budgets for the same score

We compare the two sets using the same 100 fitted logistic scores, validation samples, human histories and 1,001 thresholds. Pointwise radii are $\rho \in \{0, 0.05, 0.1, 0.2, 0.3\}$, with the existing seven deployment distances and three welfare-weight pairs. The map in (5.2) belongs to the pointwise set for $\delta \leq \rho$. At equal numerical budgets this set is contained

in the Wasserstein set: the difference is how displacement may be distributed across observations.

Average and per-observation budgets. The maximum-probability rule replaces 18.68%, 5.61% and 0.148% of doctors at pointwise radii 0.05, 0.1 and 0.2, respectively, and retains everyone at 0.3. At 0.2, only 47 of the 100 replications have replacements. The Wasserstein budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$ in this comparison all retain everyone. For $(\phi, \eta) = (2, 1)$, pointwise replacement yields mean gains over retention of 0.0275 (Monte Carlo standard error 0.0009) at $(\rho, \delta) = (0.05, 0.05)$ and 0.0052 (0.0003) at $(0.1, 0.1)$. Both deployment distributions lie on the boundary of their respective population ambiguity sets. Figure 3 shows the full distance grid; Appendix B.3 reports the numerical entries.

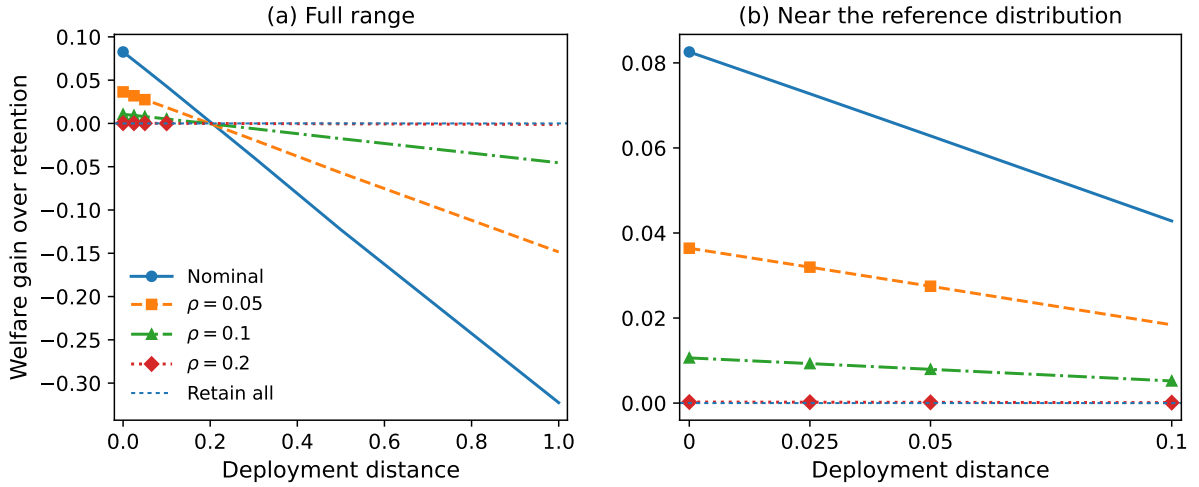


Figure 3: Replacement welfare with a bound on each observation’s displacement.

Notes: The maximum-posterior-probability rule, with $(\phi, \eta) = (2, 1)$. Curves give mean gains over retaining all doctors across the same 100 replications. Panel (a) shows all seven evaluated distances; panel (b) enlarges $0 \leq \delta \leq 0.1$. Markers identify evaluated distributions within the population ambiguity set, $\delta \leq \rho$; the same colors and line styles identify policies in both panels. The zero-radius curve is nominal replacement under either set. Pointwise $\rho = 0.3$ and all positive Wasserstein budgets in the comparison coincide with the retention line. Table 6 reports means and Monte Carlo standard errors.

Individual losses remain possible with estimated probabilities. At the two boundaries above, mean harmful fractions among replaced doctors are 0.74% and 4.04%. At $(\rho, \delta) = (0.2, 0.1)$ this fraction is 11.52% across the 47 replications with replacements, while the mean harmful share of the full doctor group is 0.016%. Five replications there have a negative aggregate gain. These results evaluate finite-sample policies; the welfare bounds retain their estimation errors. At the larger distances 0.5 and 1, every remaining replacement is harmful. Partial replacement has positive mean gains at the evaluated positive distances within the set, while larger shifts bring losses.

Choosing within the posterior feasible set. We compare the original maximum-probability threshold $\hat{c}^q$ with the welfare-based threshold $\hat{c}^W$ in (2.15), using decision

weights $(2, 1)$. The two choices replace exactly the same doctors. Table 2 shows that the welfare choice exchanges some posterior probability for a higher empirical welfare lower bound. At $\rho = 0.05$, the mean selected probability falls from 0.9774 to 0.9564, while the welfare lower bound rises from 0.9003 to 0.9207. Mean $\hat{K}_n$ falls from 0.0131 to 0.0092. The paired welfare increase at $\delta = 0.05$ is 4.15×10^{-3} (Monte Carlo standard error 0.18×10^{-3}), raising the mean gain over retention to 0.0316.

Table 2: Threshold preferences on the same pointwise feasible set.

ρ	Replaced (%)	Posterior probability		Welfare lower bound		$10^3 \hat{K}_n$	
		$\hat{c}^q$	$\hat{c}^W$	$\hat{c}^q$	$\hat{c}^W$	$\hat{c}^q$	$\hat{c}^W$
0	41.51	0.9824	0.9570	0.9335	0.9605	32.73	21.34
0.05	18.68	0.9774	0.9564	0.9003	0.9207	13.06	9.15
0.1	5.61	0.9732	0.9558	0.8574	0.8736	3.58	2.64
0.2	0.15	0.9661	0.9548	0.7660	0.7764	0.08	0.07
0.3	0.00	—	—	—	—	0.00	0.00

Notes: Both preferences use $\tau = 0.95$ and the same replacement set. Welfare quantities and the welfare-based choice use weights $(2, 1)$. Probability and welfare-lower-bound means first average over replaced doctors, then over replications with replacements: 100 at $\rho = 0, 0.05, 0.1$, 47 at 0.2, and none at 0.3. Replacement rates and $\hat{K}_n$ use all 100 replications and the full 500-doctor group. Zero-radius policies coincide across sets; positive Wasserstein budgets in this comparison retain everyone. At $\rho = \delta = 0.05$, the paired difference in deployment welfare between $\hat{c}^W$ and $\hat{c}^q$ is 4.15 (Monte Carlo standard error 0.18), in units of 10^{-3} . Table 6, panel B, reports the differences over the full grid; the data supplement retains unrounded values.

The mean welfare difference favors the welfare choice at every evaluated distance for each radius with replacements. At $(\rho, \delta) = (0.2, 1)$, seven replications have small negative differences. In the same-score comparison grid, every positive Wasserstein budget leaves both threshold choices with empty feasible sets. The comparison therefore isolates threshold preference among the doctors whose replacement is feasible, with the error costs specified only for the welfare choice.

6 An illustration on cross-state income data

The ACSIncome task predicts annual income above \$50,000 from demographic and employment covariates (Ding et al., 2021). California, New York and Texas supply the reference records; five other states supply evaluation records. We fit a logistic regression (LR) and a 100-tree random forest on 48,000 records, estimate the machine frontier on 6,000 separate records, and use another 6,000 as history for a fixed group of 200 synthetic decision makers. Each posterior therefore uses about 30 cases. The policies are selected from reference information before target evaluation. Appendix C.1 gives the allocation, synthetic decisions and computation.

The nominal policies replace 5 humans with LR and 10 with RF. Every positive budget in $\{0.1, 0.2, 0.3, 0.5\}$ retains everyone. LR uses exact Wasserstein components and RF uses

pointwise search with $K = 200$. At budget 0.1, their largest posterior probabilities are 0.0374 and 0.8954, respectively, below $\tau = 0.95$. Table 3 reports welfare on the held-out target records. Nominal replacement raises aggregate welfare in four states and lowers it in North Dakota; positive-budget retention forgoes the former gains and avoids the latter loss.

Table 3: Cross-state outcomes of the reference-based policies.

State	Marginal W_1	Target ranking AUC		All-human W	Nominal-policy gain	
		LR	RF		LR	RF
WV	1.635	0.8173	0.8404	1.0357	+0.00228	+0.00940
MS	1.624	0.8225	0.8440	1.0703	+0.00614	+0.00767
NM	1.423	0.8396	0.8659	1.0981	+0.00391	+0.01909
ND	1.668	0.7996	0.8335	1.0442	-0.00407	-0.00377
HI	2.033	0.8148	0.8491	1.0119	+0.00981	+0.01173

Notes: Welfare averages $2\text{TPR} - \text{FPR}$ equally across decision makers; gains subtract all-human welfare. AUC denotes the area under the conventional ROC curve. The nominal LR and RF policies replace 5 and 10 of the same 200 decision makers, respectively. Every positive budget in $\{0.1, 0.2, 0.3, 0.5\}$ retains everyone, so its gain is zero in every state. Distances use the specified 1,200-record samples. These are descriptive outcomes from one fixed split and synthetic roster, not means over repeated simulations.

The illustration separates two practical considerations. First, descriptive marginal feature distances range from 1.423 to 2.033, while the fixed policies have small gains or losses. These distances describe covariate differences; their relationship to a label-preserving budget also depends on class proportions and the common reference law. Human conditional rates in this synthetic exercise may change across states, so its results describe reference-based policies on those target records.

Second, search effort matters for the RF frontier. At $\rho = 0.5$, using the first 1,000 proposals rather than the first 200 of the same sequence lowers its area from 0.3937 to 0.2968. All six search sizes retain everyone. The exact empirical pointwise rule on this candidate grid also retains everyone because its posterior rectangles are contained in the sampled rectangles, even though the exact frontier values remain unknown. In the separate age-shift exercise, both nominal policies remain above all-human welfare at every evaluated displacement. Appendix C reports every budget, search prefix and age shift.

7 Conclusion

A replacement decision made before deployment must account for uncertainty about the machine’s future error rates as well as uncertainty about the human’s historical performance. The robust frontier makes this comparison threshold by threshold. Under stable human error rates, it provides a way to retain humans whose replacement cannot meet the specified posterior safety requirement.

The welfare consequences depend on both the replacement set and the chosen thresholds. Frontier estimation and posterior concentration control possible harm relative to retaining the human. A comparison with a deployment matched-FPR benchmark additionally includes the welfare difference between two machine thresholds, even when both policies replace the same person. The posterior criterion ranks thresholds by evidence of dominance, independently of the welfare weights. A welfare-based choice on the same feasible set changes this ranking and weakly reduces the threshold term in the bound.

The numerical comparisons hold the fitted score fixed and show why the form of the deployment restriction matters. Average budgets $\varepsilon \in \{0.05, 0.1, 0.2, 0.3\}$ lead to retention in this design, whereas per-observation limits of 0.05 and 0.1 support partial replacement and positive mean gains at the evaluated set boundaries. Beyond the protected range, losses return. Within a fixed posterior feasible set, welfare-based threshold choice improves the mean deployment outcomes in this design, with lower posterior probabilities of dominance. An average budget describes changes bounded in aggregate, potentially concentrated on some observations. A pointwise budget requires a justified upper bound on each observation's displacement. These restrictions should reflect the deployment changes the decision maker considers plausible. The secondary preference then determines how to choose among thresholds that meet the posterior requirement.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) for language polishing. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- Audibert, J.-Y. and Tsybakov, A. B. (2007). Fast learning rates for plug-in classifiers. *Annals of Statistics*, 35(2):608–633.
- Blanchet, J. and Murthy, K. (2019). Quantifying distributional model risk via optimal transport. *Mathematics of Operations Research*, 44(2):565–600.
- Ding, F., Hardt, M., Miller, J. P., and Schmidt, L. (2021). Retiring adult: New datasets for fair machine learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 6478–6490.
- Duchi, J. C. and Namkoong, H. (2021). Learning models with uniform performance via distributionally robust optimization. *Annals of Statistics*, 49(3):1378–1406.
- Elliott, G. and Lieli, R. P. (2013). Predicting binary outcomes. *Journal of Econometrics*, 174(1):15–26.
- Fang, Z. and Santos, A. (2019). Inference on directionally differentiable functions. *The Review of Economic Studies*, 86(1):377–412.
- Feng, K., Hong, H., and Nekipelov, D. (2026). Statistical inference of optimal allocations I: Regularities and their implications. *Journal of Econometrics*, 254:106217.
- Feng, K., Hong, H., Tang, K., and Wang, J. (2025). Statistical tests for replacing human decision makers with algorithms. *Management Science*, 71(11):9145–9170.
- Firpo, S., Galvao, A. F., and Parker, T. (2023). Uniform inference for value functions. *Journal of Econometrics*, 235(2):1680–1699.
- Gao, R. and Kleywegt, A. J. (2023). Distributionally robust stochastic optimization with Wasserstein distance. *Mathematics of Operations Research*, 48(2):603–655.
- Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30.
- Kleinberg, J., Lakkaraju, H., Leskovec, J., Ludwig, J., and Mullainathan, S. (2018). Human decisions and machine predictions. *The Quarterly Journal of Economics*, 133(1):237–293.
- Kuhn, D., Mohajerin Esfahani, P., Nguyen, V. A., and Shafieezadeh-Abadeh, S. (2019). Wasserstein distributionally robust optimization: Theory and applications in machine learning. In *Operations Research & Management Science in the Age of Analytics*, pages 130–166. INFORMS.
- Madry, A., Makelov, A., Schmidt, L., Tsipras, D., and Vladu, A. (2017). Towards deep learning models resistant to adversarial attacks. *arXiv preprint arXiv:1706.06083*.
- Mammen, E. and Tsybakov, A. B. (1999). Smooth discrimination analysis. *Annals of Statistics*, 27(6):1808–1829.
- Mohajerin Esfahani, P. and Kuhn, D. (2018). Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1–2):115–166.

- Mullainathan, S. and Obermeyer, Z. (2022). Diagnosing physician error: A machine learning approach to low-value health care. *The Quarterly Journal of Economics*, 137(2):679–727.
- Shafieezadeh-Abadeh, S., Kuhn, D., and Mohajerin Esfahani, P. (2019). Regularization via mass transportation. *Journal of Machine Learning Research*, 20(103):1–68.
- Shafieezadeh-Abadeh, S., Mohajerin Esfahani, P., and Kuhn, D. (2015). Distributionally robust logistic regression. In *Advances in Neural Information Processing Systems*, volume 28.
- Shimodaira, H. (2000). Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244.
- Sugiyama, M. and Kawanabe, M. (2012). *Machine learning in non-stationary environments: Introduction to covariate shift adaptation*. MIT press.
- Sugiyama, M., Krauledat, M., and Müller, K.-R. (2007). Covariate shift adaptation by importance weighted cross validation. *Journal of Machine Learning Research*, 8(35):985–1005.
- van der Vaart, A. W. and Wellner, J. A. (1996). *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer, New York.
- van der Vaart, A. W. and Wellner, J. A. (2011). A local maximal inequality under uniform entropy. *Electronic Journal of Statistics*, 5:192–203.

A Proofs and additional statistical results

A.1 Transport bounds

We first state an event-probability result used to handle the strict threshold. For a nonempty set $A \subseteq \mathcal{X}$, write $d(x, A) := \inf_{a \in A} \|x - a\|_2$, and take closures relative to $\mathcal{X}$.

Lemma 1 (Event probabilities at a positive transport radius). *Let $\mathcal{X}$ be a closed subset of $\mathbb{R}^d$, let μ have a finite first moment, and let A be a nonempty Borel subset of $\mathcal{X}$. For $r > 0$,*

$$\sup_{W_1(\nu, \mu) \leq r} \nu(A) = \sup_{W_1(\nu, \mu) \leq r} \nu(\bar{A}) \quad (\text{A.1})$$

$$= \inf_{\lambda \geq 0} \left\{ \lambda r + \int (1 - \lambda d(x, A))_+ \mu(dx) \right\}. \quad (\text{A.2})$$

Proof. For a nonempty closed set F , the payoff $\mathbb{I}_F$ is bounded and upper semicontinuous, and the Euclidean cost is nonnegative, continuous, and zero only on the diagonal. The space $\mathcal{X}$ is Polish. Thus Theorem 1 and the closed-event specialization in Section 2.4 of [Blanchet and Murthy \(2019\)](#) apply. Their inner maximization is

$$\sup_{z \in \mathcal{X}} \{\mathbb{I}_F(z) - \lambda \|x - z\|_2\} = (1 - \lambda d(x, F))_+.$$

This yields (A.2) for F . We now establish (A.1) without requiring $\mathbb{I}_A$ to be upper semicontinuous.

Put $F = \bar{A}$. Fix any ν with $W_1(\nu, \mu) \leq r$ and any $t \in (0, 1)$. The mixture $\nu_t = (1 - t)\nu + t\mu$ satisfies

$$W_1(\nu_t, \mu) \leq (1 - t)W_1(\nu, \mu) \leq (1 - t)r,$$

by mixing a transport plan with the identity plan. Let $\{a_k : k \geq 1\}$ be a countable dense subset of A . For $\delta > 0$, map each $x \in F$ to the first a_k with $\|x - a_k\|_2 < \delta$, and leave points outside F unchanged. Call this map T_δ . The first-index rule is Borel measurable, and $\|T_\delta(x) - x\|_2 < \delta$ on F . The image law $\tilde{\nu} = T_{\delta\#}\nu_t$ therefore satisfies

$$W_1(\tilde{\nu}, \mu) \leq \delta + (1 - t)r, \quad \tilde{\nu}(A) \geq \nu_t(F) \geq (1 - t)\nu(F).$$

Choose $0 < \delta < tr$. Then $\tilde{\nu}$ is feasible for radius r . Letting $t \downarrow 0$ and then taking the supremum over ν proves $\sup \nu(A) \geq \sup \nu(F)$. The reverse inequality follows from $A \subseteq F$. Finally $d(x, A) = d(x, F)$, so the closed-event dual gives (A.2). This argument establishes equality of values, not attainment of the supremum for A . $\square$

Proof of Proposition 1. If a coupling has finite cost under (2.6), it transports mass only within a label class. Its two class masses are therefore p and $1 - p$ under both marginals. For such a coupling, normalizing its two diagonal blocks gives couplings of (P^1, Q^1) and (P^0, Q^0) . Conversely, any pair of these conditional couplings can be combined with weights p and $1 - p$. Taking infima over the two blocks proves (2.8). If the label marginals differ, every coupling has infinite cost.

Write $S_c = \{m > c\}$ and $N_c = \mathcal{X} \setminus S_c = \{m \leq c\}$. Because the other class can be left

unchanged, (2.8) implies

$$\begin{aligned}\underline{\beta}_\varepsilon(c; P) &= \inf_{W_1(\nu, P^1) \leq \varepsilon/p} \nu(S_c), \\ \overline{\alpha}_\varepsilon(c; P) &= \sup_{W_1(\nu, P^0) \leq \varepsilon/(1-p)} \nu(S_c).\end{aligned}$$

Thus neither denominator changes during the optimization, and each conditional problem has a finite Euclidean transport cost.

For TPR, N_c is closed because m is continuous. Applying the closed-event formula to N_c and subtracting from one gives

$$\begin{aligned}\underline{\beta}_\varepsilon(c; P) &= 1 - \inf_{\lambda \geq 0} \left\{ \frac{\lambda \varepsilon}{p} + \int (1 - \lambda d(x, N_c))_+ P^1(dx) \right\} \\ &= \sup_{\lambda \geq 0} \left\{ \int \min\{1, \lambda d_c^-(x)\} P^1(dx) - \frac{\lambda \varepsilon}{p} \right\}.\end{aligned}$$

For FPR, S_c is open. Lemma 1, with $r = \varepsilon/(1-p) > 0$, gives

$$\overline{\alpha}_\varepsilon(c; P) = \inf_{\lambda \geq 0} \left\{ \frac{\lambda \varepsilon}{1-p} + \int (1 - \lambda d_c^+(x))_+ P^0(dx) \right\}.$$

Converting the conditional integrals to expectations under P yields (2.10) and (2.11).

For the multiplier restriction, write the two unnormalized objectives as

$$T(\lambda) = E_P[Y \min\{1, \lambda d_c^-(X)\}] - \lambda \varepsilon, \quad F(\lambda) = \lambda \varepsilon + E_P[(1 - Y)(1 - \lambda d_c^+(X))_+].$$

They satisfy $T(0) = 0$, $T(\lambda) \leq p - \lambda \varepsilon$, $F(0) = 1 - p$, and $F(\lambda) \geq \lambda \varepsilon$. Hence values $\lambda > p/\varepsilon$ cannot maximize T , and values $\lambda > (1-p)/\varepsilon$ cannot minimize F . Both searches can be restricted to $[0, 1/\varepsilon]$. Distances to the two nonempty regions are finite; bounded convergence makes both objectives continuous on this interval, so their extrema are attained. Replacing P by an empirical law with both classes gives the same deterministic bounds. The constant rules follow directly from their events. $\square$

Zero radius. The only law at radius zero is the reference law itself. For nonempty S_c , monotone convergence as $\lambda \rightarrow \infty$ gives

$$\inf_{\lambda \geq 0} \int (1 - \lambda d_c^+(x))_+ P^0(dx) = P^0\{d_c^+(X) = 0\} = P^0(\overline{S}_c).$$

This need not equal $P^0(S_c)$. For example, take $m(x) = (1 + e^{-x})^{-1}$, $c = 1/2$, and $P^0 = \delta_0$ on $\mathbb{R}$. The nominal strict-threshold FPR is zero, whereas the displayed zero-radius dual is one. At every positive radius the worst-case FPR is one: all negative-class mass can be moved a sufficiently small distance to the right of zero. This explains why zero-radius empirical probabilities are evaluated directly. The constant-rule cases $S_c = \emptyset$ and $S_c = \mathcal{X}$ are also evaluated directly.

A.2 Frontier attainment and posterior comparison

Lemma 2 (Attainment of the frontier and equivalent posterior criteria). *Fix P with both classes, $\varepsilon \geq 0$, a measurable score $m \in [0, 1]$, and thresholds in $[0, 1]$. Let $A(c) = \bar{\alpha}_\varepsilon(c; P)$ and $B(c) = \underline{\beta}_\varepsilon(c; P)$, using the exact probability definitions. For every $\alpha \in [0, 1]$, the threshold*

$$c_\alpha := \min\{c \in [0, 1] : A(c) \leq \alpha\}$$

exists and satisfies $g_\varepsilon(\alpha; P) = B(c_\alpha)$. For any posterior probability law Π_j on $[0, 1]^2$,

$$\sup_{c \in [0, 1]} \Pi_j\{\alpha_H \geq A(c), \beta_H \leq B(c) \mid \mathcal{D}_j\} = \sup_{\alpha \in [0, 1]} \Pi_j\{\alpha_H \geq \alpha, \beta_H \leq g_\varepsilon(\alpha; P) \mid \mathcal{D}_j\}. \quad (\text{A.3})$$

The conclusions also hold for exact empirical robust components on samples with both classes and for a finite threshold grid containing $c = 1$, with each frontier computed over its own threshold set.

Proof. Both A and B are nonincreasing. For $c_n \downarrow c$, the events $\{m > c_n\}$ increase to $\{m > c\}$. Continuity from below under each law and interchanging the two suprema give

$$\begin{aligned} A(c) &= \sup_{Q \in \mathcal{B}_\varepsilon(P)} \sup_n Q(m(X) > c_n \mid Y = 0) \\ &= \sup_n \sup_{Q \in \mathcal{B}_\varepsilon(P)} Q(m(X) > c_n \mid Y = 0) = \lim_n A(c_n). \end{aligned}$$

Thus A is right-continuous. Since $A(1) = 0$, the feasible set is nonempty and is an upper interval. Right-continuity includes its infimum in the set; if that infimum is 1, the endpoint is already feasible. This proves existence of c_α . For every feasible c , $c \geq c_\alpha$, so $B(c) \leq B(c_\alpha)$. No continuity assumption on B is needed for frontier attainment.

For any c , that threshold is feasible at $\alpha = A(c)$, hence $g_\varepsilon(A(c); P) \geq B(c)$. Its posterior rectangle is therefore contained in the frontier rectangle at $A(c)$, which proves the left side of (A.3) is no larger than the right side. Conversely, for every α the attaining threshold satisfies $A(c_\alpha) \leq \alpha$ and $B(c_\alpha) = g_\varepsilon(\alpha; P)$. Consequently,

$$\{\alpha_H \geq \alpha, \beta_H \leq g_\varepsilon(\alpha; P)\} \subseteq \{\alpha_H \geq A(c_\alpha), \beta_H \leq B(c_\alpha)\}.$$

Taking probabilities and suprema proves the reverse inequality. The same reasoning applies to an empirical reference law. On a finite grid, a smallest feasible candidate exists directly, and monotonicity proves the remaining statements. $\square$

If a maximizing frontier point exists, the two inclusions show that its implementing threshold attains the same posterior value. By contrast, the event $\{\beta_H \leq g_\varepsilon(\alpha; P)\}$ permits the threshold to vary with the posterior draw; (A.3) fixes one threshold before taking its posterior probability.

Proof of Proposition 2. Fix the reference law and posterior. Since $P \in \mathcal{B}_\varepsilon(P)$, the robust frontier satisfies $g_\varepsilon(\alpha; P) \leq g(\alpha; P)$. For each *fixed* α , therefore,

$$\{\alpha_H \geq \alpha, \beta_H \leq g_\varepsilon(\alpha; P)\} \subseteq \{\alpha_H \geq \alpha, \beta_H \leq g(\alpha; P)\}.$$

Taking posterior probabilities for this same α and then maximizing over α gives $\hat{q}_{\max, j}^\varepsilon \leq \hat{q}_{\max, j}^0$. Applying the same cutoff τ proves the claimed inclusion of replacement sets. $\square$

A.3 Logistic distances and empirical computation

Lemma 3 (Logistic distances, complexity, and finite-sample evaluation). *Let $m(x) = \{1 + \exp[-(b + x^\top \beta)]\}^{-1}$ on $\mathbb{R}^d$, with $\beta \neq 0$, and restrict c to a compact subinterval of $(0, 1)$. Then (2.19) holds. For fixed $\varepsilon > 0$ and $p_0 \in (0, 1)$, the classes $\mathcal{F}^\pm$ are bounded, pointwise measurable VC-subgraph classes and satisfy Assumption 1(ii).*

For an empirical law with $n_1, n_0 > 0$, put $n = n_1 + n_0$. For any nonnegative finite distances $d = (d_1, \dots, d_{n_y})$, define

$$V(d, \varepsilon) = \max \left\{ \frac{1}{n} \sum_{i=1}^{n_y} u_i : 0 \leq u_i \leq 1, \quad \frac{1}{n} \sum_{i=1}^{n_y} d_i u_i \leq \varepsilon \right\}.$$

At $\varepsilon > 0$ the empirical components equal

$$\widehat{\beta}_\varepsilon(c) = 1 - \frac{V((d_c^-(X_i))_{Y_i=1}, \varepsilon)}{n_1/n}, \quad \widehat{\alpha}_\varepsilon(c) = \frac{V((d_c^+(X_i))_{Y_i=0}, \varepsilon)}{n_0/n}.$$

Sorting distances and filling the budget in increasing order evaluates V , with a fraction of the last mass used when necessary.

Proof. We first compute the distances and then establish complexity and finite-sample evaluation.

Distances. The positive region is $\{z(x) > k_c\}$. If $z(x) > k_c$, projection on the hyperplane $z = k_c$ moves x by $(z(x) - k_c)\beta/\|\beta\|_2^2$, with length $(z(x) - k_c)/\|\beta\|_2$. Cauchy–Schwarz shows that no shorter displacement reaches $z \leq k_c$. Points already in that set have distance zero. The distance to $z > k_c$ is obtained by the reverse projection and an arbitrarily small additional displacement when needed. This proves both distances without changing the strict threshold convention.

Complexity. Let $a = \lambda/\|\beta\|_2$ and $\ell(v) = \min\{1, \max\{0, v\}\}$. The unnormalized class functions are

$$y\ell\{az(x) - ak_c\}, \quad (1 - y)\ell\{1 + az(x) - ak_c\}.$$

For $0 \leq u < 1$, the subgraph condition $u < \ell(az + d)$ is exactly $u < az + d$. For $u < 0$ or $u \geq 1$, membership is fixed independently of (a, d) . Thus variable subgraph membership is determined by a half-plane in the two coordinates (z, u) ; the class has finite VC dimension. Multiplication by a label indicator only partitions the domain into an active and an inactive class, and multiplication by $1/p_0$ or $1/(1 - p_0)$ rescales the subgraph coordinate. These operations retain finite VC dimension. For either class $\mathcal{F}$ with constant envelope M , [van der Vaart and Wellner \(1996, Theorem 2.6.7, p. 141\)](#), with $r = 2$, gives

$$N(uM, \mathcal{F}, L_2(H)) \leq K_0 V(\mathcal{F}) (16e)^{V(\mathcal{F})} u^{-2\{V(\mathcal{F})-1\}}, \quad 0 < u < 1.$$

Here $V(\mathcal{F})$ is the finite VC-index of the subgraphs, one more than their VC dimension, and K_0 is universal. This bound holds for every probability law H and implies Assumption 1(ii). Pointwise continuity in (λ, c) on the compact parameter domain provides a countable dense approximating family, establishing pointwise measurability. The envelopes are $1/p_0$ and $1/(1 - p_0)$.

Finite-mass optimization. The finite linear program defining V is feasible and bounded. Its scalar dual is

$$V(d, \varepsilon) = \min_{\lambda \geq 0} \left\{ \lambda \varepsilon + \frac{1}{n} \sum_{i=1}^{n_y} (1 - \lambda d_i)_+ \right\}.$$

Using $\min\{1, v\} = 1 - (1 - v)_+$ in the TPR dual, and the FPR dual itself, gives the two displayed component formulas. The program spends budget on a unit of mass at cost d_i . Exchanging expenditure from a larger positive d_i to a smaller one can only increase the transported mass, so an optimum fills the sorted distances in order; zero-distance masses are included first. The scalar objective is piecewise linear, with knots at $1/d_i$ for positive d_i . Proposition 1 permits restriction to $[0, 1/\varepsilon]$, giving the stated finite evaluation rule. Equation (A.1) justifies the open-event supremum. $\square$

A.4 Uniform estimation of the robust frontier

Proof. Write $A(c) = \bar{\alpha}_\varepsilon(c; P_0)$, $B(c) = \underline{\beta}_\varepsilon(c; P_0)$, and use hats for their empirical versions. All suprema in component errors below are over $\mathcal{C}$. The argument is conditional on the independent training sample.

Step 1: uniform control of the dual criteria. By Proposition 1, both population and sample criteria use $\Lambda = [0, 1/\varepsilon]$. For a bounded envelope M and the entropy bound in Assumption 1(ii),

$$\int_0^1 \sqrt{1 + v \log(K/u)} du < \infty.$$

The uniform-entropy maximal inequality (van der Vaart and Wellner, 2011, equation (1.1)) therefore gives

$$\sup_{f \in \mathcal{F}^+ \cup \mathcal{F}^-} |(\hat{P}_n - P_0)f| = O_p(n^{-1/2}).$$

Its measurability and bounded-envelope requirements are supplied by Assumption 1(ii). The two classes form a finite union, so their entropy integrals remain finite.

Also $\hat{p} - p_0 = O_p(n^{-1/2})$. On the event that $\hat{p}$ is bounded away from zero and one, which has probability tending to one, changing the denominators from $p_0, 1 - p_0$ to $\hat{p}, 1 - \hat{p}$ changes each criterion uniformly by $O_p(n^{-1/2})$. This includes its budget term because $\lambda \varepsilon \leq 1$. For example, the TPR numerator $\hat{P}_n[Y \min\{1, \lambda d_c^-\}] - \lambda \varepsilon$ has absolute value at most one. The FPR numerator is at most two. Thus both empirical dual criteria converge uniformly at $O_p(n^{-1/2})$. Samples missing a class have exponentially small probability and do not affect this conclusion; a bounded convention can be assigned on that event.

Step 2: optimize on the same domain. The operations $f \mapsto \sup_\lambda f$ and $f \mapsto \inf_\lambda f$ are 1-Lipschitz in the sup norm. Hence, writing

$$\delta_{\alpha,n} = \sup_c |\hat{A}(c) - A(c)|, \quad \delta_{\beta,n} = \sup_c |\hat{B}(c) - B(c)|,$$

Step 1 gives the component rate in part (i).

Step 3: bracket the inverse, including empirical jumps. Suppose $\delta_{\alpha,n} < b_C$. For every $\alpha \in [\alpha_l, \alpha_u]$, define c_α as the population inverse at level α , and $\widehat{c}_\alpha$ as the smallest feasible empirical threshold. The strict endpoint inequalities imply that these thresholds lie in $\mathcal{C}$. They also exclude every $c < c_l$ as infeasible; every $c > c_u$ is weakly dominated by the feasible candidate c_u . Thus restricting to $\mathcal{C}$ does not change either frontier on the interval under study. Lemma 2 supplies empirical attainment even if the sample components jump.

If $c < c_{\alpha+\delta_{\alpha,n}}$, then $A(c) > \alpha + \delta_{\alpha,n}$, hence $\widehat{A}(c) > \alpha$. Conversely, $c_{\alpha-\delta_{\alpha,n}}$ is empirically feasible. Therefore

$$c_{\alpha+\delta_{\alpha,n}} \leq \widehat{c}_\alpha \leq c_{\alpha-\delta_{\alpha,n}}. \quad (\text{A.4})$$

The derivative bound on the population map yields

$$\sup_{\alpha \in [\alpha_l, \alpha_u]} |\widehat{c}_\alpha - c_\alpha| \leq \delta_{\alpha,n} / \underline{\kappa}.$$

Step 4: bracket the frontier. The inclusion of feasible threshold sets gives the useful direct bound

$$g_\varepsilon(\alpha - \delta_{\alpha,n}; P_0) - \delta_{\beta,n} \leq \widehat{g}_\varepsilon(\alpha) \leq g_\varepsilon(\alpha + \delta_{\alpha,n}; P_0) + \delta_{\beta,n}. \quad (\text{A.5})$$

The population inverse and B are Lipschitz on the regular region, so its frontier has constant at most $L_\beta / \underline{\kappa}$. Hence

$$\sup_{\alpha \in [\alpha_l, \alpha_u]} |\widehat{g}_\varepsilon(\alpha) - g_\varepsilon(\alpha; P_0)| \leq \delta_{\beta,n} + \frac{L_\beta}{\underline{\kappa}} \delta_{\alpha,n}.$$

Together with part (i) and $\Pr(\delta_{\alpha,n} < b_C) \rightarrow 1$, this proves part (ii). The statement concerns exact components and continuous threshold search. Approximate components contribute their uniform numerical error; a finite grid also contributes its frontier approximation error, as specified in Section 3.2. $\square$

A.5 Functional differentiation and its limiting distribution

Theorem 1 controls the size of frontier estimation error. A functional limit additionally requires the first-order expansion to be uniform in the FPR index. Pointwise differentiation of an optimized value is not sufficient for this step (Firpo et al., 2023). We state a sufficient condition on the population multiplier sets and verify it for a smooth-score example below.

For $K = \Lambda \times \mathcal{C}$, let

$$f_{\lambda,c}^+(x, y) = y \min\{1, \lambda d_c^-(x)\}, \quad f_{\lambda,c}^-(x, y) = (1 - y)(1 - \lambda d_c^+(x))_+.$$

Represent a reference law by its moments

$$\mathcal{M}(P) = (PY, (Pf_{\lambda,c}^+)_K, (Pf_{\lambda,c}^-)_K) \in \mathbb{D} := \mathbb{R} \times \ell^\infty(K)^2, \quad (\text{A.6})$$

with the maximum of the three coordinate norms. The map to the robust frontier is defined on moment arrays generated by the reference laws in Proposition 1, with $0 < PY < 1$. For differentiation the limiting directions belong to $\mathbb{T} = \mathbb{R} \times C(K)^2$, where $C(K)$ denotes continuous functions on K . Thus an admissible sequence satisfies

$\{\mathcal{M}(P_t) - \mathcal{M}(P_0)\}/t \rightarrow h \in \mathbb{T}$ in this norm as $t \downarrow 0$; the approximating arrays need not be continuous. Define the population multiplier sets

$$\Gamma_+(c) = \arg \max_{\lambda \in \Lambda} \{P_0 f_{\lambda,c}^+ - \lambda \varepsilon\}, \quad \Gamma_-(c) = \arg \min_{\lambda \in \Lambda} \{P_0 f_{\lambda,c}^- + \lambda \varepsilon\}.$$

Theorem 7 (Functional differentiation on a regular threshold region). *Suppose Assumption 1 holds. In addition, suppose $(\lambda, c) \mapsto f_{\lambda,c}^\pm$ is continuous into $L_2(P_0)$ on K , and $c \mapsto \Gamma_\pm(c)$ is continuous in the Hausdorff distance between nonempty compact subsets of Λ . Let $A_0(c) = \bar{\alpha}_\varepsilon(c; P_0)$ and $B_0(c) = \underline{\beta}_\varepsilon(c; P_0)$. The frontier map, parameterized by (A.6), is Hadamard directionally differentiable at $\mathcal{M}(P_0)$ tangentially to $\mathbb{T}$, with values in $\ell^\infty([\alpha_l, \alpha_u])$. For $h = (h_p, h_+, h_-)$, its component derivatives are*

$$\dot{B}_h(c) = \frac{\max_{\lambda \in \Gamma_+(c)} h_+(\lambda, c) - B_0(c)h_p}{p_0}, \quad (\text{A.7})$$

$$\dot{A}_h(c) = \frac{\min_{\lambda \in \Gamma_-(c)} h_-(\lambda, c) + A_0(c)h_p}{1 - p_0}. \quad (\text{A.8})$$

They are continuous in c . Uniformly in $\alpha \in [\alpha_l, \alpha_u]$, the inverse and frontier derivatives are

$$\dot{c}_h(\alpha) = -\frac{\dot{A}_h(c_\alpha)}{A'_0(c_\alpha)}, \quad \dot{g}_h(\alpha) = \dot{B}_h(c_\alpha) - \frac{B'_0(c_\alpha)}{A'_0(c_\alpha)} \dot{A}_h(c_\alpha). \quad (\text{A.9})$$

In particular, for every admissible P_t ,

$$\left\| \frac{g_\varepsilon(\cdot; P_t) - g_\varepsilon(\cdot; P_0)}{t} - \dot{g}_h \right\|_\infty \rightarrow 0.$$

If each multiplier set is a singleton, the derivative is linear and the map is fully Hadamard differentiable tangentially to $\mathbb{T}$.

The multiplier condition allows nonunique, continuously varying optimizer sets. It supplies the uniform separation used in the following proof.

Remark 3 (Primitive conditions for the functional result). *For the logistic score on $\mathbb{R}^d$, use the normalized index $S = z(X)/\|\beta\|_2$ from Remark 1 and $a(c) = \text{logit}(c)/\|\beta\|_2$, with $\beta \neq 0$. Suppose $S \mid Y = y$ has a continuous, strictly positive density on $\mathbb{R}$ and a finite first moment. On a compact threshold interval, require*

$$\frac{\varepsilon}{p_0} < \mathbb{E}[(S - a(c))_+ \mid Y = 1], \quad \frac{\varepsilon}{1 - p_0} < \mathbb{E}[(a(c) - S)_+ \mid Y = 0] \quad \text{for every } c \in \mathcal{C}. \quad (\text{A.10})$$

Lemma 5 proves that the population multipliers are unique and continuous, the moment integrands are $L_2(P_0)$ -continuous, and both robust components are continuously differentiable with strictly negative derivatives bounded away from zero on $\mathcal{C}$. After selecting an interior FPR interval as in Assumption 1(iv), these conditions verify the functional result.

Under the same conditions, the bounded-class empirical process has a tight Gaussian limit $\mathbb{G}$ in $\mathbb{D}$, with paths in $\mathbb{T}$. Its covariance is $P_0(fg) - (P_0f)(P_0g)$ for coordinates indexed by Y and the two families $f^\pm$. The directional delta method therefore gives

$$\sqrt{n}\{\hat{g}_\varepsilon - g_\varepsilon(\cdot; P_0)\} \rightsquigarrow \dot{g}_\mathbb{G} \quad \text{in } \ell^\infty([\alpha_l, \alpha_u]). \quad (\text{A.11})$$

The proof verifies the empirical-process and tangent-space conditions before applying [Fang and Santos \(2019, Theorem 2.1\)](#). This statement is conditional on the independent training sample, with the fitted score and positive radius held fixed as the validation sample grows, and concerns the exact continuous-threshold estimator. An implemented grid or numerical approximation must have frontier error $o_p(n^{-1/2})$ relative to that estimator to share this limit.

With unique multipliers the displayed limit is Gaussian. With a nonlinear directional derivative it need not be Gaussian, and ordinary bootstrap validity does not follow from [\(A.11\)](#); see [Fang and Santos \(2019, Section 3\)](#). The limit supports smooth finite-dimensional functionals of the frontier. Estimating its law and constructing simultaneous bands are separate inference tasks.

We first establish the uniform optimization step. For nonempty compact sets $C, D \subset \Lambda$, the Hausdorff distance is

$$d_H(C, D) = \max \left\{ \sup_{x \in C} \inf_{y \in D} |x - y|, \sup_{y \in D} \inf_{x \in C} |x - y| \right\}.$$

Lemma 4 (Uniform expansion of an optimized value). *Let F be continuous on the product of compact metric spaces $\Lambda \times \mathcal{C}$, and let $V(c) = \max_{\lambda \in \Lambda} F(\lambda, c)$. Suppose $\Gamma(c) = \arg \max_{\lambda} F(\lambda, c)$ is Hausdorff-continuous. If $H_t \rightarrow H$ uniformly, with H continuous on the product and each H_t bounded, then*

$$\sup_{c \in \mathcal{C}} \left| \frac{\sup_{\lambda} \{F(\lambda, c) + tH_t(\lambda, c)\} - V(c)}{t} - \max_{\lambda \in \Gamma(c)} H(\lambda, c) \right| \rightarrow 0 \quad (t \downarrow 0).$$

The derivative is continuous in c and Lipschitz in H under the uniform norm. The corresponding minimum formula replaces both maxima by minima.

Proof. For any $\eta > 0$, all points satisfying $\text{dist}(\lambda, \Gamma(c)) \geq \eta$ are uniformly separated from the optimum:

$$\inf_{c \in \mathcal{C}, \lambda \in \Lambda: \text{dist}(\lambda, \Gamma(c)) \geq \eta} \{V(c) - F(\lambda, c)\} > 0 \quad (\text{A.12})$$

whenever this set is nonempty. Otherwise a convergent sequence in the compact product would have a limiting point attaining V while remaining at distance at least η from Γ . Continuity of F, V and of the distance to $\Gamma(c)$ makes this impossible. Continuity of V follows directly from uniform continuity of F .

For every c , choose a t^2 -maximizer $\lambda_t(c)$ of $F + tH_t$. No measurable selection is needed for this deterministic bound. Since $\sup_t \|H_t\|_\infty < \infty$ for small t ,

$$0 \leq V(c) - F(\lambda_t(c), c) \leq 2t\|H_t\|_\infty + t^2.$$

Equation [\(A.12\)](#) then gives $\sup_c \text{dist}(\lambda_t(c), \Gamma(c)) \rightarrow 0$. Evaluating $F + tH_t$ at a maximizer of H on $\Gamma(c)$ yields the lower bound $\max_{\Gamma(c)} H - \|H_t - H\|_\infty$ for the difference quotient. Evaluating at $\lambda_t(c)$ yields the upper bound $H_t(\lambda_t(c), c) + t$. Uniform continuity of H and the uniform distance bound make this upper bound at most $\max_{\Gamma(c)} H + o(1)$, uniformly in c . This proves the expansion. Hausdorff-continuity of Γ and uniform continuity of H prove continuity of $c \mapsto \max_{\Gamma(c)} H$. The Lipschitz statement follows from the elementary supremum inequality. Apply the result to $-F$ for minima. $\square$

Proof of Theorem 7. Let $\mathcal{M}(P_t) = \mathcal{M}(P_0) + th_t$, where $h_t \rightarrow h = (h_p, h_+, h_-)$ in $\mathbb{D}$ and $h \in \mathbb{T}$. We consider sequences in the stated domain of probability laws. The shared

multiplier interval is $\Lambda = [0, 1/\varepsilon]$ by Proposition 1. The $L_2(P_0)$ continuity in the theorem implies continuity of $P_0 f_{\lambda,c}^\pm$ on K by Cauchy–Schwarz.

1. Uniform quotient and envelope expansions. Write

$$\Psi_0^+(\lambda, c) = \frac{P_0 f_{\lambda,c}^+ - \lambda\varepsilon}{p_0}, \quad \Psi_0^-(\lambda, c) = \frac{P_0 f_{\lambda,c}^- + \lambda\varepsilon}{1 - p_0}.$$

Their versions at P_t are formed by replacing the moments and prevalence. The exact quotient identity, the denominator bound, and boundedness of h_t give, uniformly on K ,

$$\begin{aligned} \frac{\Psi_t^+ - \Psi_0^+}{t} &\longrightarrow D_h^+(\lambda, c) = \frac{h_+(\lambda, c) - \Psi_0^+(\lambda, c)h_p}{p_0}, \\ \frac{\Psi_t^- - \Psi_0^-}{t} &\longrightarrow D_h^-(\lambda, c) = \frac{h_-(\lambda, c) + \Psi_0^-(\lambda, c)h_p}{1 - p_0}. \end{aligned}$$

Both limiting functions are continuous. Lemma 4, applied separately to the maximum and minimum, therefore yields

$$\|B_t - B_0 - t\dot{B}_h\|_C = o(t), \quad \|A_t - A_0 - t\dot{A}_h\|_C = o(t), \quad (\text{A.13})$$

where $A_t = \bar{\alpha}_\varepsilon(\cdot; P_t)$ and $B_t = \underline{\beta}_\varepsilon(\cdot; P_t)$. On $\Gamma_+(c)$ the objective Ψ_0^+ equals $B_0(c)$; on $\Gamma_-(c)$, Ψ_0^- equals $A_0(c)$. Substitution gives (A.7)–(A.8). In particular, both derivatives are continuous in c .

2. Generalized inverse, including perturbed jumps. Let $c_{\alpha,t} = \min\{c \in \mathcal{C} : A_t(c) \leq \alpha\}$. Each A_t is nonincreasing and right-continuous, as in Lemma 2. The strict endpoint inequalities and the inverse sandwich used in Theorem 1 imply

$$\sup_\alpha |c_{\alpha,t} - c_\alpha| = O(t),$$

and keep all these points in the interior of $\mathcal{C}$ for small t . Let $R_t = A_t - A_0 - t\dot{A}_h$; by (A.13), $\|R_t\|_\infty = o(t)$. The function $A_0 + t\dot{A}_h$ is continuous. Thus every jump of A_t in this interior region is at most $2\|R_t\|_\infty = o(t)$. Minimality and right-continuity imply

$$A_t(c_{\alpha,t}) \leq \alpha \leq A_t(c_{\alpha,t}-).$$

Consequently,

$$\sup_\alpha |A_t(c_{\alpha,t}) - \alpha| = o(t). \quad (\text{A.14})$$

Since $A_0(c_\alpha) = \alpha$, uniform continuity of A'_0 and $\dot{A}_h$ gives

$$A_t(c_{\alpha,t}) - \alpha = A'_0(c_\alpha)(c_{\alpha,t} - c_\alpha) + t\dot{A}_h(c_\alpha) + o(t)$$

uniformly in α . Divide by t , use (A.14), and divide by $A'_0(c_\alpha)$, whose absolute value is at least $\underline{\kappa}$. This proves the inverse formula in (A.9) uniformly. In particular, (A.13) and continuity of $\dot{A}_h$ establish the required moving-threshold expansion

$$\sup_\alpha \left| A_t(c_{\alpha,t}) - A_0(c_{\alpha,t}) - t\dot{A}_h(c_\alpha) \right| = o(t).$$

3. Composition and the tangent space. Lemma 2 identifies the frontier at P_t as $B_t(c_{\alpha,t})$. Hence (A.13), the inverse bound, and continuity of B'_0 and $\dot{B}_h$ give

$$g_\varepsilon(\alpha; P_t) - g_\varepsilon(\alpha; P_0) = t\dot{B}_h(c_\alpha) + B'_0(c_\alpha)(c_{\alpha,t} - c_\alpha) + o(t)$$

uniformly. Substituting the inverse derivative proves the stated frontier expansion. The derivative map is continuous (indeed Lipschitz) in h , since maximum and minimum over a fixed compact set are Lipschitz and the derivative ratios are uniformly bounded. These sequential expansions establish Hadamard directional differentiability in the norm of (A.6), tangentially to $\mathbb{T}$. Directions are limiting moment arrays in the stated uniform norm. For a signed-measure direction H , the coordinates specialize to $h_p = HY$ and $h_\pm(\lambda, c) = Hf_{\lambda,c}^\pm$ when these are admissible. Singleton multiplier sets make every displayed derivative linear.

4. Empirical limit used in (A.11). The unnormalized classes $\{f_{\lambda,c}^+\}$ and $\{f_{\lambda,c}^-\}$ inherit the bounded envelope and polynomial uniform covering bound in Assumption 1; adding Y preserves these properties. Finite-dimensional convergence of their centered empirical averages follows from the multivariate central limit theorem. For tightness, take within-class differences whose $L_2(P_0)$ norm is at most δ . Pointwise approximation by the countable subclasses, followed by bounded convergence, bounds their empirical modulus by that over countably many differences with $L_2(P_0)$ norm less than 2δ . This countable class and its squares are measurable, as required by the local inequality. Its covering bound is again polynomial and its envelope is bounded by two. The local maximal inequality in van der Vaart and Wellner (2011, Theorem 2.1), after rescaling this envelope, bounds the limiting expected empirical modulus by a constant times

$$\int_0^{2\delta} \sqrt{1 + v \log(C/u)} du,$$

which tends to zero as $\delta \downarrow 0$. The additional finite-sample term in that inequality vanishes for each fixed δ before this limit. Suprema over the countable approximating classes are measurable; outer probabilities can be used for the original processes. Compactness of K and $L_2(P_0)$ continuity make the index space totally bounded and turn this stochastic equicontinuity into tightness in the product space. The limiting Gaussian process has a version with continuous paths on K . Thus

$$\sqrt{n}\{\mathcal{M}(\hat{\mathbb{P}}_n) - \mathcal{M}(P_0)\} \rightsquigarrow \mathbb{G}, \quad \Pr(\mathbb{G} \in \mathbb{T}) = 1.$$

Samples missing a class may be assigned any fixed bounded frontier: their probability tends to zero under the class-probability condition. On the other samples the moment arrays belong to the stated domain. The map and derivative satisfy the normed-space and tangent-support conditions of Fang and Santos (2019, Theorem 2.1), which proves (A.11). Applying the ordinary chain rule to a continuously differentiable finite-dimensional functional gives the corresponding functional limit. $\square$

Lemma 5 (Nonsaturated logistic threshold regions). *The primitive conditions in Remark 3 imply the additional continuity conditions of Theorem 7. On its compact threshold interval, the two robust components are continuously differentiable with strictly negative derivatives bounded away from zero.*

Proof. Let F_y and f_y be the conditional distribution and density of S given $Y = y$. For

a threshold index $a = a(c)$, put $r_1 = \varepsilon/p_0$ and $r_0 = \varepsilon/(1 - p_0)$. Strict positivity of the densities and (A.10) imply that there are unique finite $w_1(a), w_0(a) > 0$ solving

$$\int_0^{w_1(a)} u f_1(a + u) du = r_1, \quad \int_0^{w_0(a)} u f_0(a - u) du = r_0. \quad (\text{A.15})$$

Each integral is continuous, strictly increasing in its positive upper limit, starts at zero, and tends to the corresponding expectation in (A.10). Bracketing each root by two strict inequalities and using joint continuity of the integrals proves continuity of w_y on the compact a -interval.

For $\lambda > 0$, differentiation of the conditional TPR dual objective with respect to λ gives

$$\mathbb{E}[(S - a)\mathbf{1}\{a < S < a + 1/\lambda\} \mid Y = 1] - r_1.$$

There is no atom at the moving upper cutoff. This derivative is strictly decreasing from a positive value to $-r_1$, so its zero is the unique maximizer, $\lambda_+(c) = 1/w_1(a(c))$. The FPR objective derivative is

$$r_0 - \mathbb{E}[(a - S)\mathbf{1}\{a - 1/\lambda < S < a\} \mid Y = 0],$$

strictly increasing through zero, with the unique minimizer $\lambda_-(c) = 1/w_0(a(c))$. Proposition 1 places them in the common compact interval. Their continuity verifies the Hausdorff condition. The clipped distance functions are pointwise continuous in (λ, c) and bounded; dominated convergence proves their $L_2(P_0)$ continuity, including at $\lambda = 0$.

Set $z_1(a) = a + w_1(a)$ and $z_0(a) = a - w_0(a)$. Substituting the budget equations into the dual objectives yields

$$B_0(c) = 1 - F_1(z_1(a(c))), \quad A_0(c) = 1 - F_0(z_0(a(c))).$$

To differentiate without imposing differentiability of the densities, rewrite the first budget as $\int_a^{z_1} (v - a) f_1(v) dv = r_1$ and the second as $\int_{z_0}^a (a - v) f_0(v) dv = r_0$. These integrals are continuously differentiable in their endpoints, with nonzero derivatives in z_1 and z_0 . Implicit differentiation gives

$$\begin{aligned} \frac{dB_0}{da} &= - \frac{F_1(a + w_1(a)) - F_1(a)}{w_1(a)}, \\ \frac{dA_0}{da} &= - \frac{F_0(a) - F_0(a - w_0(a))}{w_0(a)}. \end{aligned}$$

Each ratio is a continuous, strictly positive average density. It has a positive minimum and a finite maximum on the compact interval. Finally, $a'(c) = 1/(\|\beta\|_2 c(1 - c))$ is positive and bounded above and away from zero on $\mathcal{C} \subset (0, 1)$, which proves the threshold-derivative claim. Selecting an FPR interval strictly inside $A_0(\mathcal{C})$ supplies the strict endpoint inequalities required separately in Assumption 1(iv). $\square$

A.6 The human posterior and the selected threshold

Lemma 6 (Dirichlet posterior calculation and concentration). *For a fixed doctor, suppose the n_j historical pairs $(Y_i, \hat{Y}_{i,j})$ are independent with a common reference-environment law, with $p_j = \mathbb{P}(Y_i = 1) \in [p_*, 1 - p_*]$ for some $p_* \in (0, 1/2]$. Use a fixed symmetric Dirichlet*

prior with parameter $a_0 > 0$ on the four cell probabilities. Then (2.17) gives independent posterior rates and (2.18) is their exact rectangle probability. Moreover, Assumption 4 holds with, for example,

$$C_j = C(a_0) := \max\{8, \exp(a_0/4)\} + 2, \quad c_j = c(p_*) := \min\{p_*/16, p_*^2/2\}. \quad (\text{A.16})$$

Proof. Combining the multinomial likelihood with the prior gives a Dirichlet posterior. Represent it by independent Gamma variables with shapes equal to the four posterior cell parameters, normalized by their common sum. This is a representation of the Dirichlet posterior used in Feng et al. (2025). In the ratios defining FPR and TPR, the common sum cancels. The ratios use disjoint pairs of independent Gamma variables. They therefore have the independent Beta laws in (2.17), proving (2.18).

We next give a tail bound that is conditional on the observed counts. If $Z \sim \text{Beta}(a, b)$, write $s = a + b$, $\mu = a/s$, and $Z = G_1/(G_1 + G_2)$ for independent unit-scale Gamma variables. For $\mu < x < 1$, exponential Markov's inequality gives

$$\mathbb{P}(Z \geq x) \leq \inf_{0 < v < 1/(1-x)} \{1 - v(1-x)\}^{-a} (1 + vx)^{-b}.$$

The choice $v = (x - \mu)/\{x(1-x)\}$ yields $\mathbb{P}(Z \geq x) \leq \exp\{-s \text{kl}(\mu, x)\}$, where

$$\text{kl}(\mu, x) = \mu \log(\mu/x) + (1 - \mu) \log\{(1 - \mu)/(1 - x)\}.$$

For $0 < x < \mu$, the same calculation with $v < 0$ bounds $\mathbb{P}(Z \leq x)$. Since $\partial_x \text{kl}(\mu, x) = (x - \mu)/\{x(1-x)\}$ and $x(1-x) \leq 1/4$, integration from μ to x gives $\text{kl}(\mu, x) \geq 2(x - \mu)^2$ on both sides. Thus, including tails beyond the support by zero,

$$\mathbb{P}(|Z - \mu| > u) \leq 2 \exp(-2su^2), \quad u > 0. \quad (\text{A.17})$$

This bound is centered at the Beta mean, not at the true Bernoulli probability.

Let $N_{j0} = \text{FP}_j + \text{TN}_j$, $N_{j1} = \text{TP}_j + \text{FN}_j$, and $m_j = \min(N_{j0}, N_{j1})$. Conditional on the label counts, the two success counts are binomial under the stated sampling model. If $m_j > 0$, let $\tilde{\theta}_j = (\text{FP}_j/N_{j0}, \text{TP}_j/N_{j1})$, and let μ_j be the vector of posterior means. Then

$$\|\mu_j - \tilde{\theta}_j\|_\infty \leq b_j := \frac{a_0}{m_j + 2a_0}.$$

The bounded-variable inequality of Hoeffding (1963) gives

$$\mathbb{P}(\|\tilde{\theta}_j - \theta_{H,j}\|_\infty > u \mid N_{j0}, N_{j1}) \leq 4 \exp(-2m_j u^2). \quad (\text{A.18})$$

For completeness, for a Bernoulli variable V , the second derivative of $\log \mathbb{E} \exp\{v(V - \mathbb{E}V)\}$ is a tilted Bernoulli variance, bounded by $1/4$. Its value and first derivative at zero vanish, so its value is at most $v^2/8$. Exponential Markov's inequality, minimized over v , gives the one-sided bound $\exp(-2Nu^2)$ for a mean of N such variables. Applying it to both sides of both class-specific means gives (A.18).

If $t \geq 4b_j$, then on $\{\|\tilde{\theta}_j - \theta_{H,j}\|_\infty \leq t/4\}$ the posterior mean lies within $t/2$ of the truth. Applying (A.17) to the two posterior coordinates gives

$$T_j(t) \leq 4 \exp\{-(m_j + 2a_0)t^2/2\}$$

on this sampling event. Its complement has conditional probability at most $4 \exp(-m_j t^2/8)$. Therefore

$$\mathbb{E}[T_j(t) \mid N_{j0}, N_{j1}] \leq 8 \exp(-m_j t^2/8).$$

If $t < 4b_j$, use $T_j(t) \leq 1$ and

$$\frac{m_j t^2}{8} < \frac{2m_j a_0^2}{(m_j + 2a_0)^2} \leq \frac{a_0}{4}.$$

When one class is absent, $m_j = 0$ and the bound by one still applies. Hence, for all label counts and $0 < t \leq 1$,

$$\mathbb{E}[T_j(t) \mid N_{j0}, N_{j1}] \leq C_a \exp(-m_j t^2/8), \quad C_a = \max\{8, e^{a_0/4}\}. \quad (\text{A.19})$$

Finally, N_{j1} is binomial with success probability p_j . Applying the same bounded-variable inequality to the class counts yields

$$\mathbb{P}(m_j < n_j p_*/2) \leq 2 \exp(-n_j p_*^2/2).$$

On its complement use (A.19); on the exceptional event use $T_j(t) \leq 1$. It follows that

$$\begin{aligned} \mathbb{E}[T_j(t)] &\leq C_a \exp(-n_j p_*^2 t^2/16) + 2 \exp(-n_j p_*^2/2) \\ &\leq (C_a + 2) \exp\{-n_j t^2 \min(p_*/16, p_*^2/2)\}, \end{aligned}$$

where $t \leq 1$ is used in the last step. These are (A.16). $\square$

Proof of Theorem 2. Write $g = g_\varepsilon(\cdot; P_0)$ and work on $\mathcal{E}_n = \{r_n \leq \delta_0/4\}$. The posterior cube

$$K_j = \{\theta_H : \|\theta_H - \theta_{H,j}\|_\infty \leq t_0\}$$

has posterior mass $1 - T_j(t_0)$.

For a dominated doctor choose the single FPR constraint $\alpha^* = \alpha_{H,j} - t_0$, which belongs to $[\alpha_l, \alpha_u]$. By frontier attainment over the candidate set, there is one threshold c^* with

$$\widehat{\alpha}_\varepsilon(c^*) \leq \alpha^*, \quad \widehat{\beta}_\varepsilon(c^*) = \widehat{g}_\varepsilon(\alpha^*).$$

For every $\theta_H \in K_j$, the FPR constraint holds. In addition,

$$\begin{aligned} \widehat{g}_\varepsilon(\alpha^*) &\geq g(\alpha_{H,j}) - L_g t_0 - r_n \\ &\geq \beta_{H,j} + \delta_0 - L_g t_0 - r_n > \beta_{H,j} + t_0 \geq \beta_H, \end{aligned}$$

because $(1 + L_g)t_0 + r_n \leq \delta_0/2$. Thus this one threshold's rectangle contains K_j , and maximizing its posterior probability gives $\widehat{q}_{\max,j}^\varepsilon \geq 1 - T_j(t_0)$.

For a non-dominated doctor, suppose a candidate rectangle intersects K_j . The intersection implies $\widehat{\alpha}_\varepsilon(c) \leq \alpha_{H,j} + t_0$ and $\widehat{\beta}_\varepsilon(c) \geq \beta_{H,j} - t_0$. But this threshold is feasible in the frontier at $\alpha_{H,j} + t_0$, so

$$\begin{aligned} \widehat{\beta}_\varepsilon(c) &\leq \widehat{g}_\varepsilon(\alpha_{H,j} + t_0) \\ &\leq g(\alpha_{H,j}) + L_g t_0 + r_n < \beta_{H,j} - t_0, \end{aligned}$$

a contradiction. Every candidate rectangle therefore has posterior mass at most $T_j(t_0)$,

which also bounds their maximum.

For a dominated doctor to be retained on $\mathcal{E}_n$, its tail must exceed $1 - \tau$; for a non-dominated doctor to be replaced, its tail must be at least τ . Consequently all wrong decisions on $\mathcal{E}_n$ are contained in $\bigcup_j \{T_j(t_0) \geq \bar{\tau}\}$. Markov's inequality and Assumption 4 yield

$$\mathbb{P}(\text{at least one wrong decision}) \leq \mathbb{P}(\mathcal{E}_n^c) + \frac{1}{\bar{\tau}} \sum_{j=1}^J \mathbb{E}[T_j(t_0)],$$

which proves (3.7). The same tail bound gives $T_j(t_0) \rightarrow_p 0$, and hence the stated limits of the posterior maxima. No independence between the validation estimator and the doctor histories is used in these union bounds. $\square$

Proof of Theorem 3. For any selected candidate, (3.9) follows from the definitions of the two uniform component errors and the worst-case population components. For each posterior draw in the selected rectangle,

$$\widehat{\alpha}_\varepsilon(\widehat{c}_j) \leq \alpha_H, \quad \widehat{\beta}_\varepsilon(\widehat{c}_j) \geq \beta_H.$$

The component-error inequalities then hold for that draw simultaneously for all admitted Q . The entire selected rectangle, whose posterior mass is at least τ , is therefore contained in the stated relaxed-dominance event.

If $T_j(t) < \tau$, this rectangle cannot be disjoint from $\{\|\theta_H - \theta_{H,j}\|_\infty \leq t\}$: their posterior intersection has mass at least $\tau - T_j(t) > 0$. Any point in the intersection implies

$$\widehat{\alpha}_\varepsilon(\widehat{c}_j) \leq \alpha_{H,j} + t, \quad \widehat{\beta}_\varepsilon(\widehat{c}_j) \geq \beta_{H,j} - t.$$

Substitution into (3.9) proves (3.10).

If the square root in (3.11) is below one, Markov's inequality gives

$$\mathbb{P}\{\max_j T_j(t_n(\delta)) > \tau/2\} \leq \frac{2JC_0}{\tau} \exp\{-c_0 n_{\min} t_n(\delta)^2\} \leq \delta.$$

If the square root is at least one, $t_n(\delta) = 1$ and the posterior tail is identically zero because both true and posterior error-rate pairs lie in $[0, 1]^2$. On the complementary event every selected rectangle intersects the relevant cube, so the simultaneous sampling statement follows. The probability bound concerns the doctor histories; the component errors may remain random and are evaluated on the actually used machine data. $\square$

A.7 Welfare comparisons

Proof of Theorem 4 and identity (3.15). On the simultaneous event in Theorem 3, every replaced doctor satisfies

$$\begin{aligned} W_j(\widehat{c}_j; Q) - W_{H,j}(Q) &= \phi[\text{TPR}_Q(\widehat{c}_j) - \beta_{H,j}] - \eta[\text{FPR}_Q(\widehat{c}_j) - \alpha_{H,j}] \\ &\geq -\phi e_{\beta,n} - \eta e_{\alpha,n} - (\phi + \eta)t_n(\delta) \end{aligned}$$

for every admitted Q . Non-replaced doctors make no difference relative to the all-human policy. Summing this inequality proves (3.14) on the same event, whose sampling probability is at least $1 - \delta$.

For (3.15), partition the doctor indices into $\mathcal{I}_n$, $\mathcal{M}_n$, $\mathcal{A}_n$ and the complement of the two replacement sets. The first group receives the machine under both policies, the second only under the benchmark, the third only under the sample policy, and the final group receives the human under both. Their differences are exactly the three displayed sums and zero, respectively. No margin assumption is used in this identity. $\square$

Proof of Theorem 5. Population comparison. For $Q \in \mathcal{B}_\varepsilon(P_0)$, every threshold feasible for the robust FPR constraint is feasible for the same deployment FPR constraint, and its deployment TPR is at least its robust lower TPR. Thus $g_\varepsilon \leq g_Q$ and $\mathcal{R}_\varepsilon(P_0) \subseteq \mathcal{R}^*(Q)$. At a doctor replaced by both population rules,

$$\begin{aligned} W_j(c_j^*; Q) - W_j(c_{\varepsilon,j}; Q) &= \phi\{g_Q(\alpha_{H,j}) - \text{TPR}_Q(c_{\varepsilon,j})\} - \eta\{\alpha_{H,j} - \text{FPR}_Q(c_{\varepsilon,j})\} \\ &\leq \phi\{g_Q(\alpha_{H,j}) - g_\varepsilon(\alpha_{H,j}; P_0)\} \leq \phi\Delta_\varepsilon(Q). \end{aligned}$$

The FPR term has a nonpositive sign. If $j \in \mathcal{R}^*(Q) \setminus \mathcal{R}_\varepsilon(P_0)$, its forgone gain is exactly $\phi\{g_Q(\alpha_{H,j}) - \beta_{H,j}\}$, and

$$0 < g_Q(\alpha_{H,j}) - \beta_{H,j} \leq \Delta_\varepsilon(Q).$$

Apply the finite-collection margin condition and sum these two contributions. When $\Delta_\varepsilon(Q) = 0$, the sets coincide and the common-replacement difference is nonpositive. This proves (3.19) in both cases.

One posterior event for all comparisons. Let $T_j(t)$ be the posterior mass outside the true-pair cube, as in Assumption 4, and define

$$\mathcal{E}(t) := \{\max_{j \leq J} T_j(t) \leq \bar{\tau}/2\}.$$

Markov's inequality and a union bound give

$$\Pr\{\mathcal{E}(t)^c\} \leq \frac{2JC_0}{\bar{\tau}} \exp(-c_0 n_{\min} t^2) = \delta.$$

No joint independence is used. The event does not depend on Q . On this event, every median $a_{n,j}$ lies in $[\alpha_{H,j} - t, \alpha_{H,j} + t]$: a median outside that interval would put at least one half of the marginal posterior outside it, whereas $T_j(t) \leq \bar{\tau}/2 \leq 1/4$. In particular, the medians and $\alpha_{H,j} \pm t$ are all in $[\alpha_l, \alpha_u]$.

Locate extra and missed replacements. Suppose first that $g_\varepsilon(\alpha_{H,j}; P_0) - \beta_{H,j} > h_n$. At the fixed level $\alpha_* := \alpha_{H,j} - t$,

$$\widehat{g}_\varepsilon(\alpha_*) \geq g_\varepsilon(\alpha_{H,j}; P_0) - L_g t - r_n > \beta_{H,j} + t.$$

An attaining candidate has estimated FPR at most $\alpha_{H,j} - t$ and estimated TPR above $\beta_{H,j} + t$. Its one fixed rectangle contains the entire posterior cube. Hence its probability is at least $1 - T_j(t) > \tau$, and doctor j is replaced. Attainment holds for the full exact frontier and for a finite candidate set containing the all-negative rule.

Conversely, if j is replaced, its selected rectangle has probability at least $\tau > T_j(t)$. It therefore intersects the cube and satisfies

$$\widehat{\alpha}_\varepsilon(\widehat{c}_j) \leq \alpha_{H,j} + t, \quad \widehat{\beta}_\varepsilon(\widehat{c}_j) \geq \beta_{H,j} - t. \quad (\text{A.20})$$

Its estimated TPR is at most $\widehat{g}_\varepsilon(\alpha_{H,j} + t)$, which yields

$$\beta_{H,j} \leq g_\varepsilon(\alpha_{H,j}; P_0) + r_n + (1 + L_g)t = g_\varepsilon(\alpha_{H,j}; P_0) + h_n.$$

For $j \in \mathcal{A}_n$, $\beta_{H,j} \geq g_Q(\alpha_{H,j}) \geq g_\varepsilon(\alpha_{H,j}; P_0)$, so

$$0 \leq \beta_{H,j} - g_Q(\alpha_{H,j}) \leq h_n.$$

For $j \in \mathcal{M}_n$, the first implication shows

$$0 < g_Q(\alpha_{H,j}) - \beta_{H,j} \leq \Delta_\varepsilon(Q) + h_n.$$

Assumption 5 now gives (3.22). These deductions do not use a strictly positive separation δ_0 for all doctors.

Threshold differences among common replacements. Write $A_{n,j} = \widehat{\alpha}_\varepsilon(\widehat{c}_j)$ and $B_{n,j} = \widehat{\beta}_\varepsilon(\widehat{c}_j)$ within this proof. For each $j \in \mathcal{I}_n$, component control and matched-FPR attainment imply

$$W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q) \leq \phi\{g_Q(\alpha_{H,j}) - B_{n,j}\} + \eta\{A_{n,j} - \alpha_{H,j}\} + \phi e_{\beta,n} + \eta e_{\alpha,n}.$$

On $\mathcal{E}(t)$,

$$g_Q(\alpha_{H,j}) \leq \Delta_\varepsilon(Q) + g_\varepsilon(\alpha_{H,j}; P_0) \leq \Delta_\varepsilon(Q) + \widehat{g}_\varepsilon(a_{n,j}) + r_n + L_g t.$$

Also $\eta(a_{n,j} - \alpha_{H,j}) \leq \eta t$. Substitution gives

$$\begin{aligned} W_j(c_j^*; Q) - W_j(\widehat{c}_j; Q) &\leq \phi\Delta_\varepsilon(Q) + v_n \\ &\quad + \phi\{\widehat{g}_\varepsilon(a_{n,j}) - B_{n,j}\} + \eta\{A_{n,j} - a_{n,j}\}. \end{aligned}$$

Replace the final two terms by their joint positive part and sum. Since $\mathcal{I}_n \subseteq \mathcal{R}_\varepsilon^{(n)}$, the common contribution is at most

$$\widehat{\pi}_n\{\phi\Delta_\varepsilon(Q) + v_n\} + \widehat{K}_n.$$

This bound uses the actual selected threshold.

Bound the remaining contributions. For a missed replacement, matched-FPR attainment makes its gain

$$W_j(c_j^*; Q) - W_{H,j}(Q) = \phi\{g_Q(\alpha_{H,j}) - \beta_{H,j}\} \leq \phi\{\Delta_\varepsilon(Q) + h_n\}.$$

For an extra replacement, (A.20) and the component-error bounds give

$$W_{H,j}(Q) - W_j(\widehat{c}_j; Q) \leq \phi(e_{\beta,n} + t) + \eta(e_{\alpha,n} + t) = u_n.$$

Multiply by the corresponding set-size bounds and add the common contribution using (3.15). This proves (3.21). All inequalities hold on the same event $\mathcal{E}(t)$ for every Q in the ball at which the specified matched-FPR benchmark is defined. $\square$

Proposition 4 (Frontier accuracy and the welfare remainder). *Under Theorem 5, keep J , $\varepsilon > 0$, and Q fixed. Suppose $n_{\min} \rightarrow \infty$ and*

$$r_n + e_{\alpha,n} + e_{\beta,n} = O_p\left(\sqrt{\frac{\log n}{n}}\right), \quad \rho_n := \sqrt{\frac{\log n}{n}} + n_{\min}^{-1/2}.$$

Then the one-sided remainder satisfies

$$\left[\text{Regret}(\varepsilon, Q) - \widehat{K}_n - \phi \widehat{\pi}_n \Delta_\varepsilon(Q)\right]_+ = O_p(\rho_n + \{\Delta_\varepsilon(Q) + \rho_n\}^{1+\gamma_M}). \quad (\text{A.21})$$

In (3.21), the missed-replacement term is of order $O_p(\{\Delta_\varepsilon(Q) + \rho_n\}^{1+\gamma_M})$ and the extra-replacement term is of order $O_p(\rho_n^{1+\gamma_M})$.

For fixed ε and Q with $\Delta_\varepsilon(Q) > 0$, this remainder bound is generally $O_p(1)$. Theorem 5 separately retains the threshold and frontier-gap contributions. Its estimation terms depend on the fitted score, function-class complexity and distance calculations, all of which may depend on covariate dimension.

Proof. Fix an arbitrarily small failure probability $\delta > 0$. Because J and the posterior constants are fixed, the t in (3.20) is $O(n_{\min}^{-1/2})$ and satisfies its interior condition for sufficiently large samples. The assumed error rates imply $h_n, u_n, v_n = O_p(\rho_n)$. Subtract $\widehat{K}_n + \phi \widehat{\pi}_n \Delta_\varepsilon(Q)$ from (3.21), take the positive part, and use $0 \leq \widehat{\pi}_n \leq 1$. On its probability- $1 - \delta$ event the result is bounded by

$$v_n + \phi C_M \{\Delta_\varepsilon(Q) + h_n\}^{1+\gamma_M} + C_M u_n h_n^{\gamma_M}.$$

The first term is $O_p(\rho_n)$, the second is $O_p(\{\Delta_\varepsilon(Q) + \rho_n\}^{1+\gamma_M})$, and the last is $O_p(\rho_n^{1+\gamma_M})$. Since δ can be chosen arbitrarily small, combining this event with the tightness bounds for the component errors proves (A.21). $\square$

A.8 Calculations for the threshold-choice example

Proof. The positive-radius frontier. Write $b = \log\{c/(1-c)\}$ and $u = 1 - b$. For $b \in [0, 1]$ both nominal class-conditional rates are u . The conditional transport budget for either class is $r = 2\varepsilon = s^2/2$. Moving an additional mass x across a threshold from a uniform density costs $x^2/2$ when that mass is available on the relevant side. Thus, if $s \leq u \leq 1 - s$, the worst FPR is $u + s$ and the worst TPR is $u - s$. Equivalently, the distance formula in Proposition 1 gives these values by transporting the closest observations first. For the strict positive event, movement to just beyond the boundary attains the same supremum value in the limit; exact attainment of the adversarial law is not needed. When $u \geq 1 - s$ the worst FPR is one; when $u \leq s$ the worst TPR is zero. Thresholds outside the support also have worst FPR one or worst TPR zero.

For $\alpha \in [0.7, 0.9]$ and $s \in (0, 0.1)$, the smallest feasible threshold has nominal rate $u = \alpha - s$, so

$$g_{P_0}(\alpha) = \alpha, \quad g_\varepsilon(\alpha; P_0) = \alpha - 2s.$$

The threshold component maps are continuously differentiable on this interior region, with the robust FPR derivative bounded away from zero as a function of c . The score is Lipschitz, and the linear-index complexity result applies. The doctor is robustly dominated since $g_\varepsilon(0.8; P_0) - 0.2 = 0.6 - 2s > 0$. For every $Q \in \mathcal{B}_\varepsilon(P_0)$, $g_Q(0.8) \geq 0.8 - 2s$. Hence

the one-doctor margin indicator is zero below $0.6 - 2s$, and at most $t/(0.6 - 2s)$ for all larger t . This verifies the stated uniform finite-collection margin condition.

At $Q = P_0$, machine welfare is $2u - u = u$. The matched-FPR threshold gives welfare 0.8 and the robust matched-FPR threshold gives $0.8 - s$. Their replacement sets coincide and their difference is s . The human welfare is $2(0.2) - 0.8 = -0.4$; both replacements improve on retaining the doctor.

A Beta tail calculation. We use an elementary integral limit. Suppose $Z_N \sim \text{Beta}(a_N, b_N)$ with $a_N/N \rightarrow a > 0$, $b_N/N \rightarrow b > 0$, and $\theta = a/(a + b)$. For fixed $x \in (\theta, 1)$,

$$-\frac{1}{N} \log \Pr(Z_N > x) \rightarrow (a + b) \text{kl}(\theta, x), \quad (\text{A.22})$$

where $\text{kl}(\theta, x) = \theta \log(\theta/x) + (1 - \theta) \log\{(1 - \theta)/(1 - x)\}$. To verify this, express the probability as the ratio of beta integrals on $[x, 1]$ and $[0, 1]$. For large N , the log integrand divided by N is

$$f_N(z) = \frac{a_N - 1}{N} \log z + \frac{b_N - 1}{N} \log(1 - z).$$

It is strictly concave, its maximizer converges to θ , and it decreases on $[x, 1]$. The numerator is at most $\exp\{Nf_N(x)\}$ and at least $h \exp\{Nf_N(x + h)\}$ for $0 < h < 1 - x$. Its logarithm divided by N therefore tends to $f(x) = a \log x + b \log(1 - x)$ by first taking $N \rightarrow \infty$ and then $h \downarrow 0$. The same upper bound at the mode and a lower bound on an interval around θ give the denominator limit $f(\theta)$. Subtracting gives (A.22). This argument applies to any parameter sequences with the displayed limits.

The actual posterior-maximizing threshold. Take iid historical observations with $P(Y = 1) = 1/2$, $P(\hat{Y}_H = 1 \mid Y = 0) = 0.8$, and $P(\hat{Y}_H = 1 \mid Y = 1) = 0.2$. These can be generated by a label-conditional diagnostic signal whose law remains fixed across environments; no change in the human pair is required. The strong law gives

$$\frac{\text{TN}_N}{N}, \frac{\text{TP}_N}{N} \rightarrow 0.1, \quad \frac{\text{FP}_N}{N}, \frac{\text{FN}_N}{N} \rightarrow 0.4 \quad \text{almost surely.}$$

In the posterior, $Z_{0,N} := 1 - \alpha_H$ and $Z_{1,N} := \beta_H$ are independent Beta variables with respective shapes

$$(a_0 + \text{TN}_N, a_0 + \text{FP}_N), \quad (a_0 + \text{TP}_N, a_0 + \text{FN}_N).$$

Let $F_{0,N}, F_{1,N}$ be their CDFs. On $s < u < 1 - s$ the exact posterior objective is

$$q_N(u) = F_{0,N}(1 - u - s)F_{1,N}(u - s).$$

It is continuous and zero at the two endpoints; outside this interval it is zero. Thus a maximum exists. Put $x_0 = 0.5 - s > 0.2$. At $u = 0.5$,

$$1 - q_N(0.5) \leq \Pr(Z_{0,N} > x_0) + \Pr(Z_{1,N} > x_0).$$

For any fixed $d \in (0, x_0 - 0.2)$, uniformly over $u \geq 0.5 + d$,

$$1 - q_N(u) \geq \Pr(Z_{0,N} > x_0 - d),$$

and uniformly over $u \leq 0.5 - d$ the analogous lower bound uses $Z_{1,N}$. Conditional on an almost-sure history path, (A.22) applies with $(a, b) = (0.1, 0.4)$ to both variables. Since $\text{kl}(0.2, x)$ is strictly increasing for $x > 0.2$, the lower bounds outside the d -neighborhood are eventually strictly larger than the upper bound at its center. Every maximizer therefore lies within that neighborhood for all sufficiently large N . Applying this argument to a countable sequence $d \downarrow 0$ proves $\hat{u}_N \rightarrow 0.5$ almost surely. The center's probability tends to one, so replacement occurs eventually at every fixed $\tau < 1$.

Posterior concentration also yields $a_{N,1} \rightarrow 0.8$. Near its limit the exact robust frontier is $a_{N,1} - 2s$, while the selected component pair is $(\hat{u}_N + s, \hat{u}_N - s)$. Consequently,

$$\hat{K}_N = [2\{a_{N,1} - 2s - (\hat{u}_N - s)\} + (\hat{u}_N + s - a_{N,1})]_+ \rightarrow 0.3 - s.$$

The deployed welfare tends to 0.5, giving the claimed signed comparison limit 0.3. The calculation isolates threshold choice with a known reference frontier. $\square$

A.9 The local-perturbation extension

We first verify the attaining selections used in the proposition. This step preserves the strict threshold and the deterministic perturbation model.

Lemma 7 (Measurable local extrema). *The selections in Definition 6 exist in either of the following cases:*

- (i) $\mathcal{X}$ is closed in $\mathbb{R}^d$ and $m : \mathcal{X} \rightarrow [0, 1]$ is continuous;
- (ii) $\mathcal{X} = \mathbb{R}^d$ and m is constant on each cell of a finite axis-aligned rectangular partition, whose open and closed faces are specified.

A fixed finite ensemble of axis-aligned trees has the partition in (ii), obtained by intersecting its leaf cells.

Proof. In (i), $K_\rho(x)$ is nonempty and compact, so the minimum and maximum exist. For any real a , the set $\{x : \underline{m}_\rho(x) \leq a\}$ is closed: if $x_n \rightarrow x$ and $z_n \in K_\rho(x_n)$ have $m(z_n) \leq a$, the z_n are bounded and have a subsequence converging to an admissible z with $m(z) \leq a$. Thus the minimum value is Borel measurable; the maximum is handled by $-m$.

Here is a direct selection construction. For a closed bounded cube D , let $v_D(x)$ be the minimum of m on $K_\rho(x) \cap D$, with value $+\infty$ if this set is empty. The same subsequence argument shows that v_D is Borel measurable. The set of x whose minimizing points intersect D is exactly $\{x : v_D(x) = \underline{m}_\rho(x)\}$, hence is Borel. Enumerate closed unit cubes covering $\mathbb{R}^d$ and select the first that intersects the minimizing set. Subdivide that cube into its finitely many closed dyadic children and again select the first intersecting child, repeating this construction. Each choice is measurable. The nested intersections with the compact minimizing set are nonempty and have diameters tending to zero. The cube centers therefore converge measurably to an attaining point. Apply the same construction to $-m$ for the maximum.

In (ii), fix a nonempty cell D and one anchor $z_D \in D$. The projection $\pi_D(x)$ onto its closed rectangular closure is continuous. A point of D is reachable within radius ρ exactly when

$$\|x - \pi_D(x)\| < \rho, \quad \text{or} \quad \|x - \pi_D(x)\| \leq \rho \text{ and } \pi_D(x) \in D.$$

Indeed, equality of the distance to a closed convex set can be attained only by its unique projection. Under the strict inequality, the points $\pi_D(x) + 2^{-k}(z_D - \pi_D(x))$ lie in D and eventually in $K_\rho(x)$. The reachability set is Borel. Choose the projection when admissible; otherwise choose the first such k . This gives a measurable reaching point for each reachable cell. At least one cell is reachable since x is admissible. Among the finitely many reachable cells, select one with the smallest score, resolving ties by a fixed cell order; do the same for the largest score. These are the desired measurable selections. $\square$

Proof of Proposition 3. For every admissible perturbation,

$$\underline{m}_\rho(X) \leq m(T_X(X, Y)) \leq \overline{m}_\rho(X).$$

Use the measurable map that sends positives to $z_\rho^-(X)$ and negatives to $z_\rho^+(X)$. It belongs to $\mathcal{B}_\rho^{\text{pt}}(P)$ and attains both component extremes. Labels and their probabilities do not change. This proves (4.5)–(4.6), including observations exactly on the score boundary.

At $\rho = 0$ the only admissible point is x , so the two scores equal m and the nominal constrained frontier is recovered. For any $Q \in \mathcal{B}_\rho^{\text{pt}}(P)$ the defining map supplies a coupling with cost at most ρ . Hence, if $\rho \leq \varepsilon$,

$$\{P\} \subseteq \mathcal{B}_\rho^{\text{pt}}(P) \subseteq \mathcal{B}_\varepsilon(P).$$

Threshold by threshold, the middle set therefore has no smaller worst TPR and no larger worst FPR than the last set, and no better components than P . Taking the maximum over the respective feasible thresholds gives (4.7). $\square$

Attainment cannot be dropped for arbitrary Borel scores. On $\mathcal{X} = [0, 1]$, set $m(0) = 1$ and $m(z) = z$ for $z > 0$. At $x = 0$, $\rho = 1$ and $c = 0$, every admissible score is strictly positive, although $\underline{m}_\rho(0) = 0$. Thus the lower-score indicator is zero while every attainable strict classification indicator is one. This example is excluded by Lemma 7; finite tree scores attain their extrema even though they are discontinuous.

For a finite proposal set containing x , its minimum is no smaller than the exact minimum, its maximum is no larger than the exact maximum, and both bracket $m(x)$. Taking class frequencies and then the same candidate frontier proves the sampled-frontier ordering stated in Section 4.1.

Proof of Theorem 6. We condition throughout on the independent training sample.

Component estimation and feasibility. Write $p_y = P_0(Y = y) > 0$ and $\hat{p}_y = N_y/n$. The classes

$$\{\mathbb{I}(Y = y)\mathbb{I}(Z_y(X) \leq c) : c \in [0, 1]\}, \quad y = 0, 1,$$

are nested indicator classes with bounded envelopes and finite VC dimension. Approximating c by decreasing rational thresholds gives pointwise measurability, also at atoms. Their polynomial covering bound has a finite uniform entropy integral. Applying the maximal inequality in van der Vaart and Wellner (2011, eq. (1.1)) gives a uniform $O_p(n^{-1/2})$ bound for their empirical averages. The class probabilities have the same rate. On $\hat{p}_y > 0$, the identity

$$\hat{F}_y(c) - F_y(c) = \frac{(\hat{\mathbb{P}}_n - P_0)[\mathbb{I}(Y = y)\mathbb{I}(Z_y(X) \leq c)] - F_y(c)(\hat{p}_y - p_y)}{\hat{p}_y}$$

gives $d_{y,n} = O_p(n^{-1/2})$. For these rate statements assign a fixed CDF on a missing-class sample, whose probability is $p_0^n + (1 - p_0)^n$. The implementation reports a missing conditional rate rather than using this asymptotic convention.

Both population and empirical components are right-continuous and nonincreasing in c . The smallest feasible threshold exists, since $c = 1$ is feasible, and maximizes the nonincreasing TPR component. If a population threshold is feasible at $\alpha - d_{0,n}$, it is empirically feasible at α . Conversely, an empirically feasible threshold at α is population-feasible at $\alpha + d_{0,n}$. The TPR components differ by at most $d_{1,n}$. Applying these two inclusions proves (4.13), without any density assumption.

Continuous score laws. The interval buffers and $f_0 \geq \kappa$ put the population and empirical inverse thresholds in C with probability tending to one. There,

$$\tilde{g}_\rho(\alpha; P_0) = 1 - F_1(c_\alpha), \quad \left| \frac{d}{d\alpha} \tilde{g}_\rho(\alpha; P_0) \right| = \frac{f_1(c_\alpha)}{f_0(c_\alpha)} \leq L_1/\kappa.$$

Use this Lipschitz bound in (4.13) to obtain the stated uniform rate.

For differentiability, let $F_{y,t} = F_y + th_{y,t}$ be CDFs with $\|h_{y,t} - h_y\|_{\infty, C} \rightarrow 0$, where h_y is continuous on C , and let $t \downarrow 0$. All inverses for $\alpha \in I$ lie in C for small t . The monotonicity sandwich and the lower density bound first give

$$\sup_{\alpha \in I} |c_{\alpha,t} - c_\alpha| = O(t), \quad c_{\alpha,t} = \inf\{c : F_{0,t}(c) \geq 1 - \alpha\}.$$

The perturbed CDF can have jumps. However, $F_0 + th_0$ is continuous on C and is within $o(t)$ of $F_{0,t}$, so its largest relevant jump is $o(t)$. Since

$$F_{0,t}(c_{\alpha,t}-) \leq 1 - \alpha \leq F_{0,t}(c_{\alpha,t}),$$

the binding residual is $o(t)$ uniformly in α . Uniform Taylor expansion of F_0 and uniform continuity of h_0 now imply

$$0 = f_0(c_\alpha)(c_{\alpha,t} - c_\alpha) + th_0(c_\alpha) + o(t).$$

Thus $(c_{\alpha,t} - c_\alpha)/t \rightarrow -h_0(c_\alpha)/f_0(c_\alpha)$ uniformly. Expanding $1 - F_{1,t}(c_{\alpha,t})$ in the same way proves (4.14). The derivative is linear and continuous in (h_0, h_1) , with norm bounded by $1 + L_1/\kappa$. This establishes Hadamard differentiability on the domain of CDF pairs, locally endowed with the uniform norm on C , tangentially to its continuous directions. The result for P is the composition with its two conditional-score CDFs, whose normalization is the quotient displayed above.

Fixed finite supports. Every $c \in [0, 1]$ has the same strict-threshold indicators as some candidate in $\{0, s_1, \dots, s_K, 1\}$. Let $d = \text{dist}(I, \mathcal{A}) > 0$. When $\max_k |\hat{A}_k - A_k| < d/2$, each candidate's feasibility agrees with its population feasibility simultaneously for every $\alpha \in I$. The first feasible index $k(\alpha)$ is therefore the same, and monotonicity of the TPR component gives

$$\hat{g}_\rho(\alpha) - \tilde{g}_\rho(\alpha; P_0) = \hat{B}_{k(\alpha)} - B_{k(\alpha)}.$$

Its supremum is at most $d_{1,n}$. The probability of the displayed feasibility event tends to one by part (i), proving the rate. For arbitrary sufficiently small perturbations of the

fixed-support conditional mass vectors the same feasibility argument holds. Each B_k is a linear sum of positive-class masses, so the last identity gives (4.15) uniformly on I . This is differentiability relative to the fixed supports, not to perturbations that insert new score values. $\square$

The boundary qualification in part (iii) is necessary. Consider a fixed one-dimensional tree score $m(x) = 1/4$ for $x \leq 0$ and $m(x) = 3/4$ for $x > 0$, with $\rho = 1/4$. In each class let X take values -1 and 1 , with

$$P_0(X = 1 \mid Y = 0) = a, \quad P_0(X = 1 \mid Y = 1) = b, \quad 0 < a, b < 1.$$

The admissible balls cannot cross the split, so the exact lower and upper scores equal m at every observed support point. The available component pairs are $(1, 1)$, (a, b) , $(0, 0)$, and

$$\tilde{g}_\rho(\alpha; P_0) = \begin{cases} 0, & 0 \leq \alpha < a, \\ b, & a \leq \alpha < 1, \\ 1, & \alpha = 1. \end{cases}$$

If $a = 1/2$, the event $\hat{a} > a$ has probability tending to $1/2$ under iid validation sampling with $0 < p_0 < 1$. On this event $\hat{g}_\rho(a) = 0$, whereas $\tilde{g}_\rho(a; P_0) = b$. Thus the frontier is not even pointwise consistent at $\alpha = a$. Deterministically, changing the negative-class mass from a to $a + t$, for any $t > 0$, changes the frontier at a from b to zero although the two conditional CDFs change by at most t . This also rules out a finite directional derivative there. The example has a positive pointwise radius and a finite tree, and does not rely on failure to compute the extrema.

Strict increase alone is likewise insufficient in the continuous case. Take conditional extremum-score CDFs on $[0, 1]$ given by $F_0(c) = 1/2 + 4(c - 1/2)^3$ and $F_1(c) = c$. They can be realized at any fixed $\rho > 0$ with $m(x) = \text{logistic}(x)$: if V_y has CDF F_y , set $X \mid Y = 0$ to have the law of $\text{logit}(V_0) - \rho$, and $X \mid Y = 1$ that of $\text{logit}(V_1) + \rho$. Both CDFs are continuous and strictly increasing, but $f_0(1/2) = 0$. The perturbed CDF $F_{0,t} = (1 - t)F_0 + t\mathbb{I}\{c \geq 1/4\}$ differs from F_0 by at most t . At $\alpha = 1/2$, its inverse shifts by $-[t/\{8(1 - t)\}]^{1/3}$ for sufficiently small t , and the frontier shifts by the opposite amount. There is no root- n transfer based only on a uniform root- n CDF error at such a point. The lower density condition in part (ii) excludes this instability.

A.10 Replacement using confidence bounds

A simultaneous bound on threshold-specific performance can be combined with the same fixed-threshold posterior rule. Let $U_{\alpha,n}(c), L_{\beta,n}(c) \in [0, 1]$ be bounds on the candidates $\mathcal{C}_n$ actually searched. To make their conservativeness relative to the plug-in rule explicit, impose

$$U_{\alpha,n}(c) \geq \hat{\alpha}_\varepsilon(c), \quad L_{\beta,n}(c) \leq \hat{\beta}_\varepsilon(c). \quad (\text{A.23})$$

Any given one-sided bands can be enlarged in these directions by taking a maximum or minimum with the plug-in estimates; this preserves their coverage. Define the fixed-threshold probability

$$\hat{q}_j^L(c) = \Pi_j\{\alpha_H \geq U_{\alpha,n}(c), \beta_H \leq L_{\beta,n}(c) \mid \mathcal{D}_j\}. \quad (\text{A.24})$$

The protected rule replaces doctor j when a maximizing candidate $\widehat{c}_j^L$ satisfies $\widehat{q}_j^L(\widehat{c}_j^L) \geq \tau$, and deploys that candidate. As for the baseline rule, a finite grid and a fixed tie rule ensure that this choice exists and is measurable.

Proposition 5 (Conditional guarantee for replacement using confidence bounds). *Maintain the stable human-comparison condition and suppose (A.23) holds. Suppose the simultaneous coverage event*

$$\mathcal{E}_n^L = \{\overline{\alpha}_\varepsilon(c; P_0) \leq U_{\alpha,n}(c), \quad \underline{\beta}_\varepsilon(c; P_0) \geq L_{\beta,n}(c) \text{ for every } c \in \mathcal{C}_n\}$$

has probability at least $1 - \gamma$, for $\gamma \in (0, 1)$. Then:

- (i) the protected replacement set is a subset of the plug-in replacement set computed on the same candidates and at the same τ ;
- (ii) on $\mathcal{E}_n^L$, each protected replacement has posterior probability at least τ of FPR/TPR dominance by its selected fixed threshold simultaneously for all $Q \in \mathcal{B}_\varepsilon(P_0)$;
- (iii) under the separation, interior-FPR, frontier-Lipschitz and posterior-concentration conditions of Theorem 2, the protected rule is consistent if

$$\sup_{c \in \mathcal{C}_n} \{|U_{\alpha,n}(c) - \overline{\alpha}_\varepsilon(c; P_0)| + |L_{\beta,n}(c) - \underline{\beta}_\varepsilon(c; P_0)|\} = o_p(1),$$

and the candidate sets contain, or approximate with vanishing component error, each of the fixed dominating thresholds used for dominated doctors in that theorem.

The proposition is conditional on valid simultaneous component bounds. It separates their frequentist coverage level $1 - \gamma$ from the posterior decision level τ ; it does not convert either level into an unconditional true-dominance probability. Theorem 7 provides a limit distribution on its stated regular regions, not a construction of these bounds. Implementing this extension requires calibrated simultaneous component bounds in addition to the plug-in quantities in Definition 3.

Proof. For every candidate c , (A.23) gives inclusion of the protected posterior rectangle in the plug-in rectangle. Consequently, $\widehat{q}_j^L(c) \leq \widehat{q}_j^\varepsilon(c)$ pointwise; maximizing on the same candidate set proves (i). This nesting is a pathwise statement and does not require the coverage event.

On $\mathcal{E}_n^L$, a parameter pair in the rectangle of the selected $\widehat{c}_j^L$ satisfies

$$\text{FPR}_Q(\widehat{c}_j^L) \leq \overline{\alpha}_\varepsilon(\widehat{c}_j^L; P_0) \leq U_{\alpha,n}(\widehat{c}_j^L) \leq \alpha_H,$$

$$\text{TPR}_Q(\widehat{c}_j^L) \geq \underline{\beta}_\varepsilon(\widehat{c}_j^L; P_0) \geq L_{\beta,n}(\widehat{c}_j^L) \geq \beta_H$$

for every admitted Q . The posterior probability of this rectangle is at least τ by selection, proving (ii). All comparisons use the same deployed threshold. The conclusion is a posterior statement on the stated frequentist coverage event.

For (iii), let $\delta_n^L = o_p(1)$ bound the two uniform component discrepancies. For a dominated doctor take the fixed dominating threshold from Theorem 2, or its candidate approximation. Its population FPR is strictly below $\alpha_{H,j}$ and its TPR strictly above $\beta_{H,j}$. For sufficiently small δ_n^L and approximation error, its protected rectangle contains a fixed neighborhood of the true human pair. The posterior mass of this neighborhood tends to

one by the mean posterior-tail condition and Markov's inequality. Hence its protected probability tends to one.

For a non-dominated doctor, suppose a protected rectangle at a candidate c intersects $\{\|\theta_H - \theta_{H,j}\|_\infty \leq t\}$. Then

$$\bar{\alpha}_\varepsilon(c; P_0) \leq \alpha_{H,j} + t + \delta_n^L, \quad \underline{\beta}_\varepsilon(c; P_0) \geq \beta_{H,j} - t - \delta_n^L.$$

Thus $\beta_{H,j} \leq g_\varepsilon(\alpha_{H,j} + t + \delta_n^L; P_0) + t + \delta_n^L$. Choose a fixed sufficiently small $t > 0$ so that the FPR argument is interior and $(1 + L_g)(t + \delta_n^L) < \delta_0$ with probability tending to one. Frontier Lipschitzness makes the last inequality incompatible with non-dominance and separation. Every candidate rectangle is therefore contained in the complement of this neighborhood. Its posterior probability tends to zero uniformly over the candidates. Since J is fixed, these statements prove simultaneous consistency. $\square$

B Population evaluation in the smooth-score experiment

The reference index in (5.1) has distribution $R \sim N(b_0, v)$, where $v = \|\beta\|_2^2 + \sigma_u^2$. Write $\pi(r) = (1 + e^{-r})^{-1}$ and $p_0 = E[\pi(R)]$. Let Φ denote the standard normal CDF. A doctor's conditional decision probability given $R = r$ is

$$H_j(r) = \Phi\left(\frac{\pi(r) - c_{H,j}}{\sigma_\nu}\right).$$

Hence the human reference rates, identical at every δ , are

$$\alpha_{H,j} = \frac{E[(1 - \pi(R))H_j(R)]}{1 - p_0}, \quad \beta_{H,j} = \frac{E[\pi(R)H_j(R)]}{p_0}. \quad (\text{B.1})$$

Condition on the fitted score $m(x) = \{1 + \exp[-(a + x^\top w)]\}^{-1}$, with $w \neq 0$. For $M = a + Z^\top w$, the joint normal law of (M, R) gives

$$M \mid R = r \sim N(\mu(r), s_M^2), \quad \mu(r) = a + \frac{w^\top \beta}{v}(r - b_0), \quad s_M^2 = \|w\|_2^2 - \frac{(w^\top \beta)^2}{v} > 0.$$

Positivity follows from $\sigma_u > 0$ and Cauchy-Schwarz. For an interior score threshold c , put $t = \log\{c/(1 - c)\}$. Since the deployed index is $M + (1 - 2Y)\delta w_1$,

$$\begin{aligned} \text{FPR}_{Q_\delta}(c) &= \frac{E[(1 - \pi(R))\Phi\{(\mu(R) + \delta w_1 - t)/s_M\}]}{1 - p_0}, \\ \text{TPR}_{Q_\delta}(c) &= \frac{E[\pi(R)\Phi\{(\mu(R) - \delta w_1 - t)/s_M\}]}{p_0}. \end{aligned} \quad (\text{B.2})$$

The endpoints $c = 0, 1$ are evaluated directly. All expectations in (B.1)–(B.2) are one-dimensional.

Let $A_0(t)$ denote the reference negative-class survival probability of M . Solve $A_0(t_{0,j}) = \alpha_{H,j}$. Conditional normal noise has positive variance and the label weights are positive, so the conditional index densities are positive on the real line; the matching threshold exists and is unique. The deployment matched-FPR index cutoff is $t_{0,j} + \delta w_1$. Its TPR is

the reference positive-class survival probability evaluated at $t_{0,j} + 2\delta w_1$. This cutoff is used for the benchmark only; it does not replace the posterior-selected candidate.

The numerical reference uses 1,024-point Gauss–Hermite integration and compares it with 512-point integration at all evaluated human rates, selected machine thresholds and matching cutoffs. The maximum difference must be at most 5×10^{-7} , and the matched-FPR probability residual must be at most 10^{-10} . These are numerical convergence checks, not rigorous integration-error certificates. Counts of welfare gains and benchmark dominance comparisons within the numerical tolerance of zero are retained with the results. In a replication with no replacements, the harmful fraction among replaced doctors is undefined and is reported as missing; the fraction among all doctors is zero.

B.1 Alternative welfare weights

The posterior rule does not use (ϕ, η) , so the same decisions and thresholds underlie all three welfare comparisons. Table 4 reports the paired difference between each of the four positive-radius policies in Table 1 and the standard policy at every prespecified deployment distance. Each entry uses all 100 replications. No averaging across radii is involved: the four positive-radius policies coincide doctor by doctor because they retain everyone.

Table 4: Robust-minus-standard welfare under the prespecified weights.

δ	$(\phi, \eta) = (1, 1)$	$(2, 1)$	$(5, 1)$
0	-0.0507 (0.0010)	-0.0826 (0.0017)	-0.1781 (0.0039)
0.025	-0.0445 (0.0009)	-0.0727 (0.0015)	-0.1575 (0.0034)
0.05	-0.0382 (0.0008)	-0.0628 (0.0013)	-0.1368 (0.0030)
0.1	-0.0254 (0.0005)	-0.0428 (0.0009)	-0.0951 (0.0023)
0.3	0.0275 (0.0006)	0.0392 (0.0010)	0.0745 (0.0022)
0.5	0.0822 (0.0017)	0.1228 (0.0027)	0.2446 (0.0055)
1	0.2168 (0.0045)	0.3226 (0.0068)	0.6398 (0.0135)

Notes: Means over 100 independent replications; parentheses give the standard deviation of the within-replication difference divided by $\sqrt{100}$. Every positive radius in $\{0.05, 0.1, 0.2, 0.3\}$ gives these same welfare differences. A positive entry favors retention under the robust policy; a negative entry measures the foregone gain from standard replacement. The main comparison $(2, 1)$ is included for reference.

B.2 The finer-radius sensitivity

The entries below use the same replications and comparisons as Figure 2.

Table 5: Radius sensitivity: replacement and welfare relative to retaining all doctors.

ε	Replaced (%)	Welfare minus all-human welfare, $(\phi, \eta) = (2, 1)$						
		$\delta = 0$	0.025	0.05	0.1	0.3	0.5	1
0	41.51 (0.88)	0.08257 (0.00173)	0.07274 (0.00153)	0.06284 (0.00133)	0.04283 (0.00093)	-0.03923 (0.00098)	-0.12279 (0.00266)	-0.32258 (0.00676)
0.0001	22.79 (0.70)	0.04628 (0.00138)	0.04083 (0.00122)	0.03532 (0.00105)	0.02419 (0.00073)	-0.02166 (0.00074)	-0.06860 (0.00214)	-0.18150 (0.00550)
0.0003	12.98 (0.49)	0.02666 (0.00099)	0.02354 (0.00087)	0.02039 (0.00076)	0.01401 (0.00052)	-0.01233 (0.00051)	-0.03940 (0.00151)	-0.10477 (0.00394)
0.001	3.18 (0.17)	0.00659 (0.00035)	0.00582 (0.00031)	0.00505 (0.00027)	0.00347 (0.00018)	-0.00307 (0.00018)	-0.00983 (0.00054)	-0.02627 (0.00143)
0.003	0.14 (0.02)	0.00028 (0.00004)	0.00025 (0.00003)	0.00022 (0.00003)	0.00015 (0.00002)	-0.00013 (0.00002)	-0.00043 (0.00006)	-0.00115 (0.00016)
0.01	0.00 (0.00)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
0.05	0.00 (0.00)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)

Notes: Means over the same 100 replications as Table 1; Monte Carlo standard errors are in parentheses. Welfare differences are paired with the all-human policy within replications. At $\varepsilon = 0.003$, 44 replications replace at least one doctor, but aggregate means include all 100. For radii 0.0001–0.01, only $\delta = 0$ lies in the corresponding population ambiguity set among the evaluated points. The complete 147-cell supplement reports all three prespecified welfare weights, paired standard-policy differences, conditional harm measures and effective counts. Standard errors describe variability across the same 100 replications.

B.3 The same-score budget and threshold comparisons

Tables 6–7 report welfare and conditional harmful fractions at all seven deployment distances and five pointwise radii. In this same-score grid, the positive Wasserstein budgets yield retention under both preferences; zero-budget outcomes coincide across sets. The data supplement gives the other evaluation weights, full-group losses and effective counts.

Table 6: Welfare under the two pointwise threshold preferences.

$\rho \backslash \delta$	0	0.025	0.05	0.1	0.3	0.5	1
<i>A. Maximum posterior probability: gain over retaining everyone</i>							
0	0.08257 [†] (0.00173)	0.07274 (0.00153)	0.06284 (0.00133)	0.04283 (0.00093)	-0.03923 (0.00098)	-0.12279 (0.00266)	-0.32258 (0.00676)
0.05	0.03642 [†] (0.00123)	0.03196 [†] (0.00108)	0.02748 [†] (0.00093)	0.01841 (0.00063)	-0.01890 (0.00071)	-0.05701 (0.00200)	-0.14851 (0.00509)
0.1	0.01063 [†] (0.00053)	0.00929 [†] (0.00047)	0.00794 [†] (0.00040)	0.00520 [†] (0.00026)	-0.00606 (0.00032)	-0.01760 (0.00090)	-0.04543 (0.00230)
0.2	0.00025 [†] (0.00003)	0.00021 [†] (0.00003)	0.00018 [†] (0.00002)	0.00010 [†] (0.00002)	-0.00019 (0.00003)	-0.00049 (0.00007)	-0.00122 (0.00017)
0.3	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
<i>B. Welfare choice minus maximum-probability choice</i>							
0	0.01196 [†] (0.00037)	0.01193 (0.00037)	0.01191 (0.00036)	0.01183 (0.00036)	0.01123 (0.00033)	0.01023 (0.00029)	0.00646 (0.00017)
0.05	0.00416 [†] (0.00018)	0.00415 [†] (0.00018)	0.00415 [†] (0.00018)	0.00413 (0.00018)	0.00397 (0.00016)	0.00366 (0.00015)	0.00243 (0.00009)
0.1	0.00101 [†] (0.00006)	0.00101 [†] (0.00006)	0.00101 [†] (0.00006)	0.00101 [†] (0.00006)	0.00098 (0.00006)	0.00092 (0.00005)	0.00064 (0.00003)
0.2	0.00002 [†] (0.00000)	0.00002 [†] (0.00000)	0.00002 [†] (0.00000)	0.00002 [†] (0.00000)	0.00002 (0.00000)	0.00002 (0.00000)	0.00001 (0.00000)
0.3	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 [†] (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)

Notes: Means and Monte Carlo standard errors (in parentheses) over the same 100 saved replications. Differences are paired within replication. Welfare weights are (2, 1). A dagger marks a population-set point, $\delta \leq \rho$. Aggregate means include all 100 replications, including those with no replacements. Wasserstein policies at the positive budgets in this comparison have zero gain over retaining everyone and zero difference between threshold choices. The numerical supplement contains all three evaluation-weight pairs for the maximum-probability rule, full-group harmful shares, loss magnitudes, and the complete paired comparisons.

Table 7: Harmful fractions under the two pointwise threshold preferences.

$\rho \backslash \delta$	0	0.025	0.05	0.1	0.3	0.5	1
<i>A. Maximum posterior probability</i>							
0	0.17 [†] (0.03)	0.31 (0.05)	0.70 (0.08)	2.82 (0.20)	99.94 (0.02)	100.00 (0.00)	100.00 (0.00)
0.05	0.24 [†] (0.06)	0.40 [†] (0.09)	0.74 [†] (0.13)	2.99 (0.27)	99.98 (0.02)	100.00 (0.00)	100.00 (0.00)
0.1	0.13 [†] (0.06)	0.25 [†] (0.08)	0.65 [†] (0.14)	4.04 [†] (0.59)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)
0.2	0.00 [†] (0.00)	0.00 [†] (0.00)	3.37 [†] (2.28)	11.52 [†] (4.37)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)
0.3	— —	— —	— —	— —	— —	— —	— —
<i>B. Welfare choice</i>							
0	0.03 [†] (0.01)	0.06 (0.02)	0.17 (0.03)	0.86 (0.09)	98.71 (0.13)	100.00 (0.00)	100.00 (0.00)
0.05	0.10 [†] (0.03)	0.12 [†] (0.04)	0.31 [†] (0.07)	1.11 (0.14)	99.66 (0.06)	100.00 (0.00)	100.00 (0.00)
0.1	0.04 [†] (0.04)	0.07 [†] (0.05)	0.16 [†] (0.08)	1.60 [†] (0.31)	99.92 (0.04)	100.00 (0.00)	100.00 (0.00)
0.2	0.00 [†] (0.00)	0.00 [†] (0.00)	2.13 [†] (2.13)	8.69 [†] (3.77)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)
0.3	— —	— —	— —	— —	— —	— —	— —

Notes: Mean fractions among replaced doctors, in percent, with Monte Carlo standard errors in parentheses. Welfare weights are (2, 1). A dagger marks a population-set point, $\delta \leq \rho$. Conditional means use 100 replications at radii 0, 0.05 and 0.1, and the 47 replications with replacements at 0.2. At 0.3 nobody is replaced, so the fractions are undefined. The numerical supplement contains all three evaluation-weight pairs for the maximum-probability rule, full-group harmful shares, loss magnitudes, and the complete paired comparisons.

C Additional ACS summaries

C.1 Sample construction and policy evaluation

We use the 2018 one-year records, the task’s ten original numeric features, and its eligibility filter: age above 16, income above \$100, positive hours worked, and person weight at least one. The analysis gives each retained record equal weight. A fixed sample of 20,000 eligible records from each reference state yields 60,000 records. We use 48,000 for model fitting, 6,000 to estimate the machine frontier, and the remaining 6,000 as human history. Median imputation and standardization are fitted only on the model-training sample. The predictors are logistic regression and a 100-tree random forest.

A single roster of $J = 200$ synthetic decision makers is used throughout. Their thresholds $c_{H,j}$ are drawn from $0.15 + 0.7 \text{Beta}(2, 3)$. We clip the fitted logistic proxy to $[10^{-5}, 1 - 10^{-5}]$, add U to its logit, truncate that index to $[-50, 50]$, and apply the logistic link to obtain p_{priv} . The human decision is $\mathbb{I}\{p_{\text{priv}} + \nu > c_{H,j}\}$, with $U \sim N(0, 0.15^2)$ and $\nu \sim N(0, 0.225^2)$. Reference-history counts determine the independent Beta posteriors with $a_0 = 0.01$. We select a fixed machine threshold by Definition 3, using $\tau = 0.95$, before evaluating target-state outcomes.

For the retrospective conditional-rate comparison, records are allocated to the fixed roster evenly within each observed label class, with random allocation within a class. This gives every decision maker observations from both classes. Both machines share these cases and human decisions. All eligible target-state records, subject to the 20,000-record cap, are reserved for evaluation. Synthetic human error rates can change across states, so this geographic comparison does not impose the stable-human condition of the main welfare results.

Robustness calculations and evaluation. The candidate set contains 400 equally spaced thresholds from 0.001 to 0.999, together with zero and one. Logistic regression uses the exact empirical Wasserstein calculation, whereas the forest uses the pointwise finite search in Section 4.1, with $K = 200$. Both grids are 0, 0.1, 0.2, 0.3, 0.5; ε and ρ retain their distinct average and per-record meanings. At $\rho = 0.5$, we use the first $K \in \{20, 50, 100, 200, 500, 1000\}$ proposals from the same 1,000-proposal forest search to measure sensitivity to search size. Each policy deploys the threshold used in its posterior calculation.

Descriptive feature distances compare equal-weight samples of 1,200 records from the pooled validation set and each target state, using unscaled Euclidean costs in the training-standardized coordinates. They are marginal distances, not coverage statements for the label-preserving joint set, and do not determine the budget grid. Numeric category-code perturbations describe the fitted predictor’s sensitivity rather than feasible changes in personal characteristics.

For each synthetic decision maker, evaluation welfare is twice the assigned policy’s TPR minus its FPR. We average these values equally across decision makers. Pooled-case rates are reported separately. Harm measures are finite evaluation-sample comparisons, with conditional measures left undefined when nobody is replaced. Conventional ranking AUC and the area under the deterministic candidate frontier are distinct summaries.

The age-shift exercise uses 5,000 additional California records excluded from the reference pool. It adds $\delta \in \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$ to the standardized age field read by the machine, holding labels and the unshifted human decisions fixed. The nominal and

0.5-budget policies remain those selected from reference information. This shift measures displacement from the California base; distance from the pooled reference law is separate.

C.2 Budget and search summaries

Table 8 collects the budget, search and age-shift results for the fixed illustration. The accompanying data supplement contains every policy–environment combination, decision-maker-level confusion counts, pooled-case welfare and the conditional harm measures.

Table 8: Candidate-frontier and age-shift summaries for the ACS illustration.

<i>A. Budget grid: frontier areas and maximum posterior probabilities</i>				
Budget	LR area	RF area	LR maximum q	RF maximum q
0	0.8526	0.8790	0.9850	0.9907
0.1	0.2631	0.7386	0.0374	0.8954
0.2	0.0851	0.6337	1.03×10^{-4}	0.7032
0.3	0.0155	0.5230	2.79×10^{-9}	0.4298
0.5	0.0000	0.3937	0	0.1522

<i>B. Nested forest searches at $\rho = 0.5$</i>			
K	Frontier area	Maximum q	Replaced count
20	0.5816	0.5834	0
50	0.5004	0.3855	0
100	0.4433	0.2485	0
200	0.3937	0.1522	0
500	0.3356	0.0786	0
1000	0.2968	0.0454	0

<i>C. Reserved-California age shift: nominal policies</i>			
δ	LR gain over all-human	RF gain over all-human	RF ranking AUC
0	+0.00750	+0.01316	0.8814
0.5	+0.00845	+0.01227	0.8737
1	+0.01212	+0.01360	0.8687
1.5	+0.01100	+0.01424	0.8633
2	+0.00860	+0.01258	0.8609
2.5	+0.00883	+0.01055	0.8569
3	+0.01048	+0.01022	0.8508

Notes: Panel A uses exact empirical Wasserstein components for LR and $K = 200$ feasible perturbations for RF. The two budgets have different meanings. Areas integrate the deterministic candidate frontier, not the conventional ranking ROC or a single least-favorable distribution’s ROC. Panel B holds the model, validation records and proposal sequence fixed; it supplies no bound on the unsearched region. Panel C uses the same 5,000 records and human decisions throughout, with all-human welfare 1.13206 and $(\phi, \eta) = (2, 1)$. The nominal LR ranking AUC is 0.85690 at every shift. Both 0.5-budget policies have zero gain at every shift because they retain everyone. No simulation standard errors are attached to this single split.